Learning About Presidential Power

November 4, 2026

Abstract

Presidents inform the public about their powers by using them. News organizations increasingly focus on public policy enacted by decree, which subsidizes reporting by offering a simple narrative structure. We argue this shifts public attention toward presidents and their initiatives. Using a dataset of verified coverage of executive actions in 51 newspapers, we develop new regional measures of unilateral action salience. Pairing these data with Google Trends search results for "Obama" and "Trump", we show, via a matched difference-in-differences design, that media attention to executive action strongly predicts subsequent search-engine traffic about the American presidency. Increased traffic lasts nearly one week and is similar in magnitude to the increase in searches for a musical artist following a new album release. In summary, our study reveals how the use of presidential power attracts the public's attention.

Keywords: presidency, political communication, agenda setting, public policy

Wordcount: 8,987

To hold their government accountable, people need to know who is doing the governing. Almost no one witnesses the enactment of policy, and most do not experience its direct effects. Instead, people learn about political institutions and public policy through the news (Boydstun, 2013; Soroka and Wlezien, 2022). This makes the ability to drive the news cycle potentially valuable for politicians. When politicians get their actions into the news, their preferences and accomplishments make their way to the public (Grimmer, Westwood and Messing, 2013). Information animates political contests and helps determine electoral success.

In this study, we argue the American presidency has an important, under-recognized advantage in this contest of information: the ability to take executive action. The president has long been thought to possess a loud megaphone, the so-called "bully pulpit," that can drive public attention towards favorable policies or viewpoints (e.g., Canes-Wrone, 2006; Kernell, 2007; Noble, 2024). Yet, evidence that presidential rhetoric influences public attention is hard to come by. What scholarship we have suggests presidential speech is unpersuasive (Edwards, 2006; Franco, Grimmer and Lim, 2018; Howell, Porter and Wood, 2020). But presidents can do more than talk. Executive action allows chief executives to propose and often implement changes to policy alone (Carey and Shugart, 1998; Moe and Howell, 1999). There is no shortage of research about why presidents act this way (e.g., Howell, 2003; Lowande, 2024), what its limitations are (Chiou and Rothenberg, 2017; Rudalevige, 2021; Bolton and Thrower, 2022), and what the public thinks about it (Christenson and Kriner, 2020; Reeves and Rogowski, 2021).[1] Research generally acknowledges executive action's potential effect on the news and subsequent public attention, but the reasons for such effects are not clear, and the effects themselves remain unmeasured.

We argue executive action offers presidents a means to not only change policy but enter and remain in the fast-paced American news cycle. To properly evaluate this argument, we report several innovations in measurement and research design. First, we use thousands of articles from 51 newspapers with varied geographic circulation to measure the amount of coverage of unilateral actions from the Obama and first Trump presidencies. We show that prior measures

[1]For a recent review, see Lowande and Rogowski (2021).

of unilateral action coverage, which have often relied only on newspaper articles in top national outlets, have underestimated the amount of news coverage of unilateral actions. With these data in hand, we estimate measures of the salience of executive actions in the media. These measures differ from past estimates, in that our hierarchical Bayesian Item Response Theory (IRT) model allows executive actions to matter more in some regions of the country, and less in others. These measures reveal that, consistent with research on the so-called "nationalization" of politics, salience of executive action is essentially uniform across the country (Hopkins, 2018).

We next ask whether salience in the media corresponds to salience in the *public*. We use Google Trends search traffic, which provides remarkably granular measures of public attention without some of the drawbacks of survey research (e.g., Mellon, 2013; Dancy and Fariss, 2023; Stephens-Davidowitz, 2013). It is passively collected, so our study is not limited to assessing public reactions to a particular executive action over this period. We deploy a matched difference-in-difference design to estimate the effect of salient unilateral action on Google searches for the president. Specifically, we match search traffic within a geographic region in periods in which presidents unilaterally acted, to periods without such action in the same presidential term and proximity to an election. We then compare changes in search traffic related to the president in these periods over time.

We find that executive action which is salient in the media has strong, positive effects on web-traffic. The most media-salient actions lead to a 25% relative increase in searches related to the president, and can last for up to a week. To gauge the substantive size of these effects, we deploy a similar research design to study attention spikes for non-political news. Specifically, we show executive action's attention impacts are comparable in scale to search behavior following the release of a new album by the most popular musical artists in the U.S.

We later discuss how these findings might generalize beyond the scope of our data. However, if executive action serves an agenda-setting function unique to chief executives, then there are important implications for understanding the separation of powers in democratic systems. Decree authority would not just give chief executives advantages in the production of policy. It

would give them the opportunity to move first in the contest for public attention, and perhaps to hold it or limit the opportunity for the opposition to break through. In contrast, members of legislatures often have limited tools—floor speeches, hearings, public statements, bill sponsorship, and roll call votes—to break into news cycles. The president's messaging upper-hand may have important implications for public opinion, which have gone unexamined because of sub-disciplinary boundaries in the social sciences. Our evidence suggests the president's impact on the public agenda comes by way of action. These actions are generally the purview of scholars who study "elite" institutions, while the public agenda is the purview of those who study public opinion. Our findings demand further research on how the actions of chief executives shape public understandings of presidential power, and modern democracy, more broadly.

Policymaking and Information in American Politics

Information is critical in all political interactions. Democratic republics do not work well unless citizens know what politicians stand for, what decisions politicians make, and how those decisions impact them. Two well-known, stylized facts complicate that need. First, people tend to not know much about politics. If you ask them questions about who is in charge, or what they have done, most people run out of information quickly (Kinder and Kalmoe, 2017). Second, what people believe appears to be malleable—often cued by elite actors with an interest in public beliefs—and is sometimes inconsistent with reality (e.g., Bullock et al., 2015; Iyengar et al., 2019).

These inconvenient facts run through a large academic literature about democracy and responsiveness in American politics. Since the beginning of the "behavioral revolution" in social science, scholars have argued the public does not possess sufficient information to hold its government accountable, and that most decisions are made on the basis of vague in-group and out-group distinctions (e.g., Campbell et al., 1960). Others argue that responsiveness can work just fine, because voters care about issues, and because politicians themselves believe in the possibility that they will be rewarded and punished for their policy choices (e.g., Soroka and

Wlezien, 2010; Caughey and Warshaw, 2022). A large, related literature on the media salience of public policy argues the media is the critical mechanism that makes some issues and policies more relevant than others (e.g., McCombs and Shaw, 1972; Boydstun, 2013).

Several practices common in this research area highlight the contributions of our study. Put simply, what "policy," "politician," and "media" mean in practice is vague by design. System-level analysis calls for researchers to abstract away from the behavior of particular institutions and the policies they contribute to. All heterogeneity within them, along with the messiness of rules and institutions, is usually subsumed in a single aggregate signal and response within some time window.

Our aim is different—to document and measure a particular institutional mechanism by which citizens learn about policy and politicians. Studying these mechanisms, in general, is worthwhile because some mechanisms may be more efficacious than others. Put differently, Soroka and Wlezien (2022) argue the media provide sufficiently relevant information about what politicians are up to. They show evidence that this link exists in the aggregate. But there follows an equally important question, which is whether this media link advantages some political actors over others.

Politicians are in a constant battle for public attention. Members of Congress (and the parties within which they caucus), for example, are thought to be so desperate for it that most of their behavior can be tied to the need to say who they are, what they believe, and attribute some kind of success to their efforts as a party or an individual (Mayhew, 1974; Grimmer and Stewart, 2013). Presidents have similar desires, albeit qualitatively different tools with which to pursue them.

For decades, presidency scholars have investigated presidents' efforts to hold the public's attention. But this is mostly the study of talking. Jeffrey K. Tulis (1987) showed that the acceptability of presidents publicly expressing opinions on policy took a long time to develop. Samuel Kernell (2007) argued that presidents' success appealing directly to the people could be used as a lever to bend Congresses toward their will. Brandice Canes-Wrone (2006) presumed no such persuasive power, instead, arguing that presidents merely highlighted inconsistencies

between what voters wanted, and what Congress intended to do. Even this modest conception of public appeals would advantage the presidency.

But beyond illustrative cases, it is difficult to find evidence that appeals move public opinion. George C. Edwards (2006) finds no consistent evidence that the polls follow appeals. If there is any evidence of the effect of presidential rhetoric, it is usually temporary and not one with direct implications for public opinion. Franco, Grimmer and Lim (2018) show that newscasts and social media posts temporarily respond to presidential rhetoric. Howell, Porter and Wood (2020) show experimental evidence that the public's perceptions of the president can be changed by the president's rhetoric, but again, not their opinions. Most recently, Noble (2024) has argued that presidents can alter opinions about the opposition party by going negative.

We think the behavior in question—speeches and public appeals—itself explains why evidence is often hard to come by. Speeches can be complicated and difficult for journalists to clearly summarize, if they decide to cover them at all. In addition, most of these public appeals are tied to legislative initiatives, not those within the executive branch—which is our focus. Today, the competition for the public's attention is fierce, even within politics, and politics itself is far down the list of things normal people care about (Prior, 2007). If there is a signal that voters consistently pick up, it must be stronger, more focused, and more reliably delivered than rhetoric.

The tools the president uses to ostensibly control the federal bureaucracy might be valuable for holding the attention of the public. Cooper (2014) argues that presidential directives can be used to generate favorable publicity and play "shell games" with particular constituencies. More recently, Lowande (2024) argues that executive action itself is motivated by the desire to keep up appearances, and shows it often produces favorable press coverage. But the extent to which the public receives these signals is not known. Public opinion about executive power has been studied, but it is always thought of as a potential constraint on executive action itself (e.g., Christenson and Kriner, 2020; Reeves and Rogowski, 2021; Lowande and Rogowski, 2022).

Our study, then, builds the case that executive action influences the kind of information the public consumes about American politics. In this way, executive action might give presidents

an advantage in the contest of providing voters with the minimum amount of relevant information needed for American democracy to function. Making news and reaching the public is not just about competing rhetoric, it is about action.

Executive Action and the Politics of Attention

Our goal is to explain why the issuance of executive actions ultimately shift attention toward the sitting president. We start by recalling the president's goals and strategies. Presidents have sincere preferences about policy, and want to be popular—both goals are tied to re-election, electing copartisans, and building a historical legacy. Executive action is one of many tools that serves these proximate goals. They leverage constitutional and delegated powers, along with ambiguity in statutes and their capacity as the chief executive officer of the government, to change what the government does.

Presidents know that their executive actions will be seen. Reeves and Rogowski (2021) argue the public has underlying preferences about the exercise of executive power, and so presidents avoid exceeding their authority. Christenson and Kriner (2020) argue presidents pay costs for the policy position signaled by their action, rather than the means. Both works demonstrate the public finds the narrative that presidents use executive action to circumvent gridlock very persuasive. Lowande (2024) argues that presidents' desire to be popular sometimes cuts against the preference for influencing public policy. As a result, the purpose of executive actions is often to help the president reinforce their brand as a politician. All of this suggests that presidents will avoid taking executive actions that draw negative attention (Major, 2014).

It is important to note that executive action is particularly useful for drawing media attention. The production of news is neither costless nor perfectly indexed to the substantive importance of events. In research on the "significance" of public policy, most scholars acknowledge that media attention is an imperfect measure of the "true" importance of policy (e.g., Boydstun, 2013). News stories have a set of standard components—the who, what, when, where, and how of politics—that have to be communicated efficiently, typically at an 11th grade reading level, to a public that may have little contextual information.

Executive action subsidizes the production of news in several ways. First, it provides a subject, the president, who is already the most famous politician in the world. Their motivations and role in civic life are self-contained in their title and person. Second, executive action often defines itself in ways that are helpful to reporters. The preamble of a presidential directive defines the president's position and provides context.

Executive action provides a familiar, powerful narrative about the "why" of news. It typically portrays the president taking executive action as a result of a gridlocked Congress. This narrative feeds into existing beat coverage. Curry, Lee and Oldham (2024) show that news about Congress typically emphasizes failures and negativity over success and accomplishment. News about executive action provides the perfect contrast. It is also important to remember that journalists work within space constraints. Political news cannot delve deeply into caveats, conditional logic, or contingency.

This argument has an important implication which we take to the data in the following sections. Specifically,

Media-salient executive action should increase public attention toward the sitting president.

Because presidents know the media will tend to take up their initiatives and newsworthy stories, they will tend to select policies that shine favorable light on their governance. Presidents and their advisors will help journalists write their stories. The news will be produced *because* journalists anticipate it will be consumed. The main subject of that coverage, by design, will be the sitting president. News coverage has independent effects on public attention to issues like crime, natural disasters, and unemployment (e.g., Garz and Martin, 2021; Zhao, Rosoff and John, 2019; Gramlich and Eddy, 2024). Therefore, enhanced coverage of the president should correspond to heighten public attention.

In short, executive action might confer a kind of public agenda-setting power on the president.[2] We argue, however, this simple expectation does not have self-evident implications for

[2]It is important to note that executive action is a "bundled" treatment, in the sense that there are many behaviors that feed into the ultimate news coverage that we later is associated with salience. Thus, we do not argue that speeches, press releases, and travel have no impact on media coverage or public attention, just that executive action is at least the initial catalyst.

the president's power or accountability relationship with the public. In particular, the scope and scale of this agenda-setting is important. A tool that shifts public attention temporarily in some constituencies would be qualitatively different than one that shifted attention nationally. More specifically, we investigate two parameters that might impact the ultimate importance of this tool.

Durability. The implications of attention durability are straightforward. In past work on public agenda setting, researchers have found scant evidence that politicians are able to systematically change what the public thinks is pressing or needs action on. But the lack of evidence does not imply that politicians have no influence. In this case, because of the granularity of our data, we can recover more precise estimates of how long executive action shifts public attention, if at all.

Geographic scope. Some executive action is regionally significant. Actions targeting historically black colleges and universities, for example, will be relevant where those institutions are prominent educators and employers (i.e., the Midwest and South). The creation of new national monuments will be especially relevant to the communities adjacent to those monuments. Media organizations try to find stories relevant to their readers. If attention shifts toward the president have a regional dimension, it would mean that the president might be able to target actions to particular constituencies.

On the other hand, news about executive action could be nationalized, without regional variation tied to relevance. A number of factors could lead to uniform coverage across the country. Many newspapers in our study, for instance, publish syndicated articles from the Associated Press and other national outlets. Likewise, despite regional constituencies, growing nationalization might increase the national public's demand for information about what the president is doing and the press' willingness to supply coverage.

In short, we argue that executive action presents presidents with a unique opportunity to shift public attention, the particulars of which are relevant for understanding presidents' ac-

countability relationship with the public.

Evidence

We leverage three datasets to examine how the public becomes aware of executive policymaking. The first lists executive actions taken by presidents from 1989-2021 (Lowande, 2024). The second consists of new data on the coverage of these unilateral actions across 51 newspapers. We use these data to estimate new measures of action salience that vary by geographic region. Finally, we measure public responses to that media salience using Google Trends data.

Policy Salience in the News

Testing the impact of policymaking on information-seeking behavior first requires a measure of the salience of executive actions. For every DACA or travel ban order, there are actions creating obscure commissions that receive little press coverage. We develop measures of action salience using a dataset of newspaper coverage of 1,174 unilateral actions issued by Presidents H.W. Bush, Clinton, Bush, Obama, and Trump. These actions include not only executive orders but presidential memoranda, proclamations, national security directives, and other types of policy statements issued by presidents—even those that do not include the president's signature.[3]

This dataset marks a significant improvement in describing how the news media cover unilateral actions (e.g., Mayer and Price, 2002). A team of undergraduate research assistants searched for instances of relevant coverage across 51 U.S. newspapers in the ProQuest news database. The locations of the newspapers in Figure 1 are plotted, along with the 9 geographic regions that we group papers into for purposes of calculating regional salience scores.[4] An article counted as providing coverage if it mentioned the action and attributed it to the pres-

[3]This is a random sample of approximately 68% of all unilateral actions issued between 1989 and 2021. For discussion of these actions and how they were collected, see REDACTED and REDACTED.

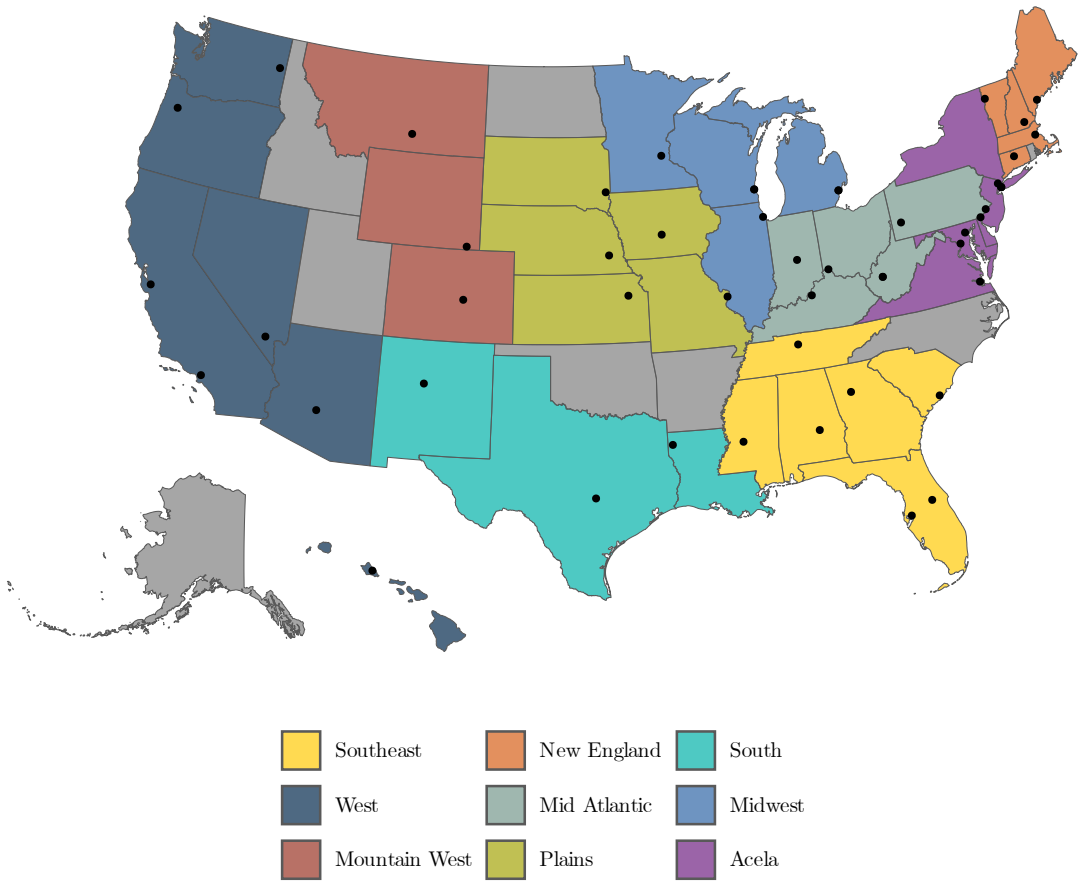
[4]Table A.1 lists the newspapers along with their locations and political affiliations. What counts as "relevant coverage" along with coding procedures, are detailed at length in the SI. See REDACTED for additional details.

ident. An article could be counted as covering an executive order if the relevant terms—e.g., "executive order" or "presidential memorandum"—were omitted. What matters is that the article provided sufficient information so that a reader could understand that a policy action took place and that it was executed by the president.

Our sweep of news coverage goes beyond examining coverage in major national outlets like the *New York Times* and *Wall Street Journal*. Executive action can vary in relevance by geographic region, so we constructed a sample of newspapers geographically representative of the United States. Our sample includes regional outlets like the *Wyoming Tribune Eagle* and *Montgomery Advertiser*. In general, we selected the newspaper with the highest circulation in each state, subject to availability in Proquest (Gentzkow, Shapiro and Sinkinson, 2014).[5]

[5]Sometimes this newspaper was not included our source database, and so the next highest newspaper was included.

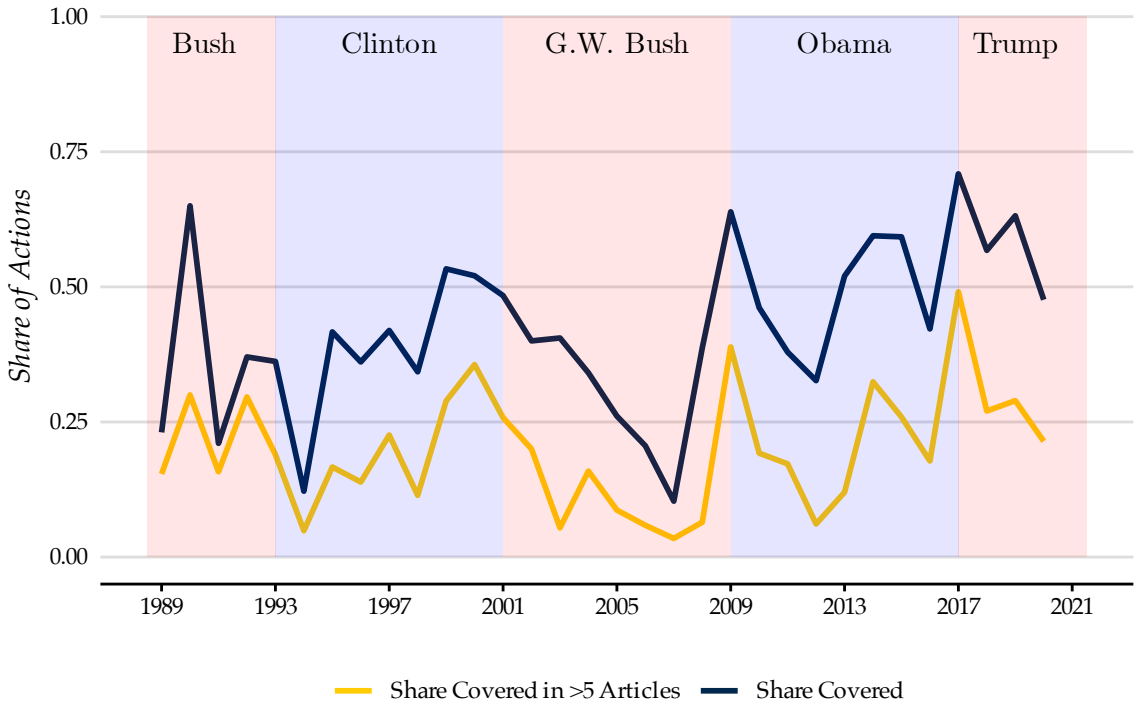
Figure 1 – Newspaper Locations and Geographic Regions. Shows the locations of the 51 newspapers in our sample within 9 geographic regions. The following states are excluded due to a lack of newspaper coverage: ID, UT, AK, ND, OK, AR, NC, and RI.



Overall, across all unilateral actions in our dataset, 51% receive mention in at least one newspaper article. This is notably higher than others have found (e.g., Christenson and Kriner, 2020), likely due to our broader cross-section of media outlets, and the fact that we did not rely on explicit mentions of directives themselves. Figure 2 shows trends in unilateral action coverage across presidential administrations. Most coverage occurs in the days and weeks immediately following the issuance of an action, and thus, Figure 2 only plots coverage in the month following the issuance of an action. This also alleviates censoring issues with actions issued later in presidents' terms. In most years, less than half of unilateral actions receive

coverage in the first month. Substantial coverage is even lower. Only about 21% of actions are covered in more than 5 newspaper articles. Overall coverage has increased during the Obama and Trump administrations. This underscores the degree to which executive action has become an important part of what Americans read about government, and more generally, the notion that media coverage tends to "reflect" rather than effect public policy (Soroka and Wlezien, 2024).

Figure 2 – Coverage of Unilateral Actions is Increasing Over Time. Shows the share of actions that received coverage at least 1 article and in more than 5 articles in the first month after issuance. Coverage is truncated to the first month to alleviate right-side censoring for actions issued late in presidents' terms.



Measure of News Attention. The degree to which an executive action is made noticeable to the public, which we call "news attention" or "salience" interchangeably, mediates the relationship between executive action and public information-seeking behavior. Although the news attention of an action may be associated with its policy impact or political significance, what is relevant to our argument is purely informational, i.e., the extent to which information about the action is made available to the public through news coverage. Newspaper coverage

is therefore an appropriate source for measuring the salience of executive actions. However, the raw number of articles covering an action is an imperfect measure of news attention, because the information conveyed by newspapers varies. An action receiving coverage in the *The New York Times* does not have the same impact on salience as coverage in *The Billings Gazette*. The former, with a national focus and a relatively large circulation and significant staff resources, is able to cover presidential policymaking with more depth and scope. In contrast, *The Billings Gazette* might have to pick and choose what it covers more closely, staying attuned to the arenas of policy most important to local readers.

Thus, following Chiou and Rothenberg (2014) and others, we use a Bayesian IRT model to recover the salience of each executive action as a latent variable. As is the case in the previous studies, the observed coverage of actions by newspapers is manifest indicators of the latent salience. Unlike Chiou and Rothenberg (2014), however, our IRT model allows for newspapers' coverage to have a varying influence on news attention. Other than the large increase in the number of newspaper "raters," our approach builds upon prior models of executive action "significance" by modeling raters as units within geographic regions. Let $j, j = 1, \ldots, J$ be the index of presidential actions, $k, k = 1, \ldots K$ be the index of newspapers, and $r, r = 1, \ldots R$ be the index of geographical regions. Each newspaper is located in a region, which is denoted by $r[k]$ and plotted in Figure 1. National newspapers are not associated with a particular region but are assumed to have national readership. The locations of the newspapers and their associated regions are plotted in Figure 1. $Y_{j,k}$ is the binary outcome of interest, denoting whether a newspaper k covered action j. We model the outcome as the latent variable Y^*_{jk}:

$$Y_{jk} = \begin{cases} 1 & \text{if} \quad Y^*_{jk} > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$Y^*_{jk} = \begin{cases} -\alpha_k + z_{j,r[k]}\beta_k + \epsilon_{jk} & \text{if newspaper } k \text{ is local} \\ -\alpha_k + w_j\beta_k + \epsilon_{jk} & \text{if newspaper } k \text{ is national} \end{cases} \tag{1}$$

where $\epsilon_{jk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.

Moreover, z_{jr} is modeled hierarchically such that

$$z_{jr} \overset{\text{indep.}}{\sim} \mathcal{N}(w_j, \omega^2). \tag{2}$$

This specification models the probability of a newspaper covering an action as a function of its regional salience z_{jr} or national salience w_j depending on whether k is local or national, a "difficulty" parameter α_k, and a "discrimination" parameter β_k.

The key departure of our model from Chiou and Rothenberg (2014) is that our model accounts for the heterogeneity of salience across regions. We define action j's national-level salience, w_j, and its region-specific salience for region r, z_{jr}, and model z_{jr} and w_j hierarchically as specified in Equation (2). This model captures the fact that the salience of presidential actions may be different across regions, but that each action also has a common level of salience shared across regions. Generally, newspaper k's coverage of action j is determined by the extent to which the action is salient in their own region, $z_{j,r[k]}$. However, newspapers with national readership cover presidential actions based on their national level of salience rather than the regional level, and therefore coverage by these newspapers carries information about w_j directly, as shown in Equation (1). In short, this specification allows us to test the extent to which news attention is nationalized, rather than assuming it is.

The hierarchical structure of Equation (2) also captures the possibility of news coverage being correlated across outlets while accounting for the fact that each newspaper still has to

decide whether to cover an action on its own. For example, the more regional newspapers reprint stories on action j from wire services, the larger w_j is estimated to be, which in turn increases the probability that other newspapers will cover action j. On the other hand, the model is flexible enough so that newspaper k's decision to cover action j informs its regional salience $z_{j,r[k]}$.

α_k is the "difficulty" parameter of newspaper k. This parameter represents newspaper k's overall level of attention to politics. Larger α_k suggest that newspaper k is less likely to cover policies in general and that an action must be relatively more salient in order to be covered by the paper. The β_k term is the "discrimination" parameter for newspaper k, which indicates the extent to which the paper's decision to cover an action depends on its latent salience. A higher value for β means that the given newspaper's coverage varies by an action's salience and thus its coverage is better able to distinguish between actions with varying levels of salience. We report these parameters in SI Section B.

To complete the model, we place the following priors on the model parameters:

$$\begin{aligned} w_j &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \\ \omega^2 &\sim Scaled-inv-\chi^2(10,1), \\ \alpha_k &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,3), \text{ and} \\ \beta_k &\overset{\text{i.i.d.}}{\sim} \mathcal{TN}_{\beta_k>0}(0,3), \end{aligned} \tag{3}$$

The prior variance of w_j is fixed to 1 for the identification of the latent scale of w_j, and the support of β_k is constrained to be positive so that the direction of w_j is identified (larger values indicating more salience).

We generate posterior draws of $\left(\left(w_j, (z_{jr})_{r=1}^{R}\right)_{j=1}^{J}, (\alpha_k, \beta_k)_{k=1}^{K}, \omega \right)$ using Stan, running three MCMC chains for 15,000 iterations (warm up = 2,500). Gelman-Rubin diagnostics provide strong evidence of convergence (Gelman and Rubin, 1992).[6] Our key result of interest are the regional salience scores z_{jr}.[7]

[6]See Figure B.1 in the Supplementary Information.
[7]The estimates of α and β for each newspaper are shown in Figures B.2 and B.3

Measuring Public Attention. With these measures in hand, we can investigate whether salient executive action corresponds to heighten public attention. We measure the public's attention to unilateral policymaking via Google Trends search data.[8] With the post-processing we describe in this section, they provide longitudinal, day-level information on the frequency of searches for phrases on Google's search engine.

There are several advantages to this measure. First and foremost, it is an outcome closely related to our research question. It represents a particular kind of mass behavior—namely, the tendency to look up information on a particular subject—in our case, the incumbent president.[9] The action itself is small and low cost, but, in the aggregate, takes stock of what is on internet users' minds. Google Trends data have been fruitfully leverage by scholars across the social sciences, to measure interest in human rights (Dancy and Fariss, 2023), predict flu epidemics (Ginsberg, Mohebbi and Patel, 2009), and the salience of the economy for American voters (Mellon, 2013).

Second, this data source does not require episodic collection, the way boutique public opinion surveys do. It can be applied to any event with contemporaneous data coverage. This allows us to compare points in time with and without executive action, similar in logic to conventional survey experiments which expose respondents to vignettes.

As Mellon (2013) notes, using Google Trends to measure the salience of topics in the public poses some research challenges. The first is that Google users might not be representative of the public, a problem Mellon shows is obviated by the widespread availability of the internet in the United States. The second is the ambiguity of terms, as searches for "economy" could be driven by the desire for news about macro-economic indicators, or about low-cost automobiles. We address this issue by ensuring our results are robust to the least ambiguous term referring to the incumbent (i.e., "Trump" and "Obama"). Finally, look-up behavior is not the same as political support. For that reason, we conclude our presentation of findings with a discussion

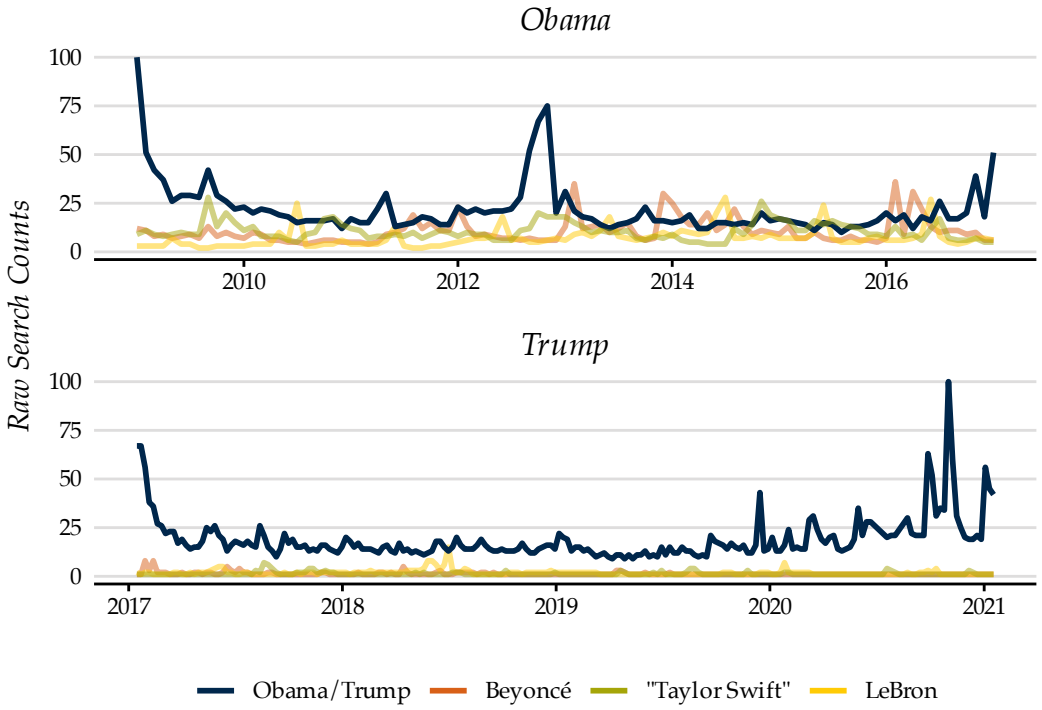
[8]These data were accessed via the 3rd party vendor SerpApi.
[9]Note, this distinguishes our measure of studies like Franco, Grimmer and Lim (2018), which study the effect of presidential action on the salience of issues, rather than the presidency itself.

of the conditions under which searches might work in presidents' favor.

We use Google Trends to measure public searches of "Obama" and "Trump" throughout both presidents' respective terms.[10] Figure 3 shows the raw search results for both "Obama" and "Trump" during their respective terms, relative to other famous individuals in the United States. Per Google Trends restrictions, searches are scaled within the range 0-100 for a given period and, given the length of Obama and Trump's terms, only available at monthly and weekly intervals, respectively. Both presidents are frequently the subject of Google searches, although the bottom facet underlines that Trump is frequently top-of-mind for many Americans. While searches for Obama are high, they often resemble search counts for other high-profile individuals, like Beyoncé Knowles-Carter, Taylor Swift, and LeBron James. Searches for Trump, in contrast, consistently outpaced searches for all of these individuals.

[10]Search results span President Obama's first and second terms and President Trump's first term. We opt to use only the president's names to ensure we are including different search phrasings. We show in Figure C.1 that searches for "Obama" and "Trump" greatly exceed searches for "President Obama" and "President Trump." However, our results are robust to using these alternative search phrases.

Figure 3 – Presidents, Especially Trump, are Frequently the Subject of Google Searches. Shows raw Google Trends search results for "Obama" and "Trump" during their respective terms relative to searches for "Beyoncé", "Taylor Swift", and "LeBron". Searches are national and on a common [0-100] scale within a presidential term. Results during Obama's term are at a monthly interval, while searches for Trump are at weekly intervals.



In addition to providing some sense of scale relative to non-political topics, Figure 3 also hints at three main issues with using Google Trends data for our analysis. First, the data is aggregated within a particular geographic area and time period. It cannot be tied directly to individual users. As a result, it does not permit inferences about individual-level behavior. Second, Google provides a normalized integer value between 0 and 100 based on the frequency of searches for the term relative to all searches over the period. As a result, it is not possible to directly compare raw search results across different periods of time or regions. Third, Google also limits day- and week-level searches to 8-month and 5-year chunks, respectively. Acquiring longitudinal search data over longer time periods is only possible at the month level.

We address the second and third challenges by imputing search results on a consistent scale. Bleher and Dimpfl (2022) provide an algorithm for knitting together overlapping intervals of

Google Trends search results. They show that two sets of search results for the same phrase and region across overlapping time periods follow a linear relationship. As such, OLS can be used to impute search results on the same scale across two different but overlapping time periods. Following the advice of Eichenauer et al. (2021), we modify the approach of Bleher and Dimpfl (2022) and incorporate week- and month-level data to better capture long-term trends in search patterns.

We start by downloading day-, week-, and month-level search data for "Obama" and "Trump." For both presidents, day-level data is downloaded in 6 month chunks. Adjacent chunks have 5 months of intersecting data. Since we only focus on Trump's first term, monthly data is not available and weekly data can be downloaded in a single chunk. We download weekly data for Obama in 3 chunks that span 4 years and overlap by 2 years. Monthly data is available for the entirety of Obama's term in a single chunk. A president has n day-level chunks of search data and, beginning with the nth and $n-1$ chunks, we start by estimating the following model over the 5 months that chunks n and $n-1$ overlap:[11]

$$Searches_{r,i,d|period_n} = \gamma + \beta Searches_{r,i,d|period_{n-1}} + \alpha \widehat{Searches}_{r,i,w} + \zeta Searches_{r,i,m} + \epsilon_{r,i,d} \quad (4)$$

Where r indexes the region of the country (defined above) or the entire country,[12] i indexes the search phrase (i.e., "Obama" or "Trump"), and d, w, and m denote, respectively, the day, week, and month. $\widehat{Searches}_{r,i,w}$ denotes search results for phrase i during the week w that includes day d. These week-level search results are estimated beforehand using a similar procedure, and are therefore already on a common scale. The final term $Searches_{r,i,m}$ denotes search results for phrase i during the month m that includes day d.

After fitting this model on the days that fall within both chunks, we estimate the predicted number of searches for the out-of-sample days from chunk $n-1$. These predicted search re-

[11] This procedure outline here is for Obama, since it is not possible to download monthly search results for Trump and weekly results are in a single chunk. See the SI for more information.

[12] We get regional search results by calculating the mean number of searches across each of the states in the region, weighted by population size.

sults, which are now on the same scale as the chunk n searches, are then appended to the search results from chunk n. Working backward in time, we then repeat the same process, using the newly predicted values to fit the same regression for the next chunk, until we have looped through all chunks in a president's term.[13] SI Section C provides additional details about this process.

Figure 4 – Google Searches for Presidents Obama and Trump. Shows Google searches for "Obama" and "Trump" during their respective terms on a common within-president scale. Search counts cannot be compared across presidents. Counts are national and calculated using the smoothing algorithm run from the most recent to the earliest chunk of search results. See Figure C.2 for results calculated using the algorithm run from the earliest to the most recent period.

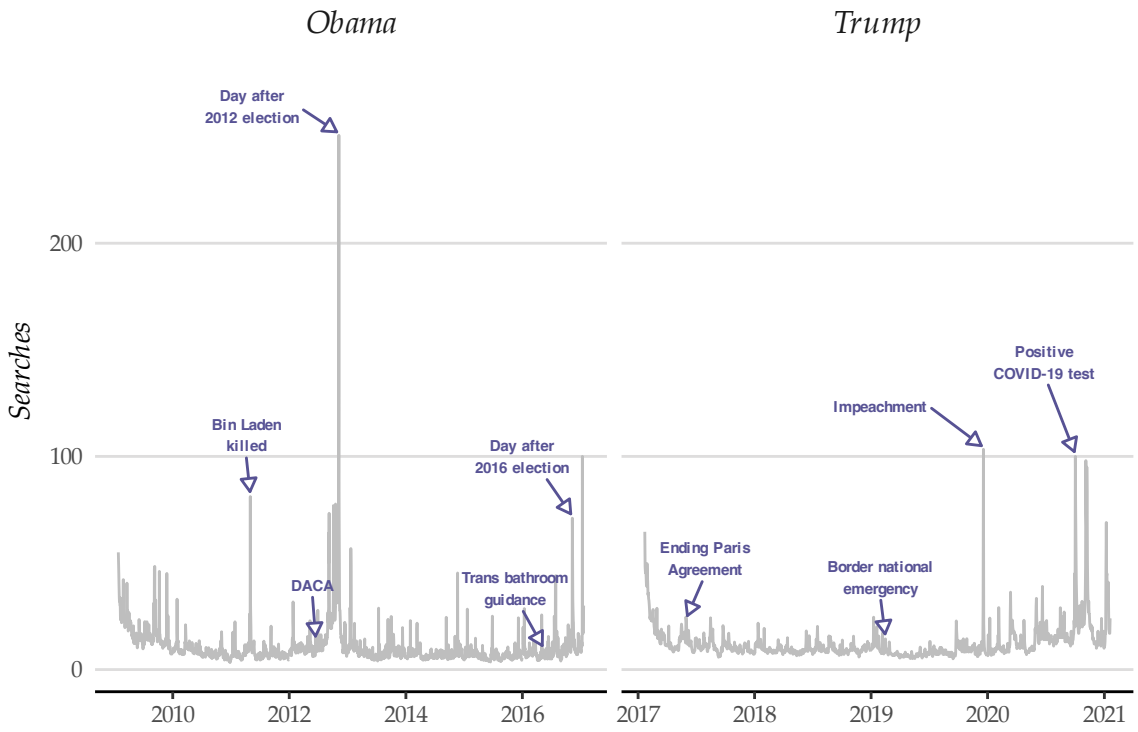


Figure 4 shows the results of our standardization procedure. Each facet plots the day-level

[13]We opt to impute backwards rather that forward from the earliest period to the most recent because of apparent changes in the quality of Google Trends data over time. (Namely, preliminary state-level searches for "President Obama" resulted in frequent zeroes in less-populated states during Obama's first term.) However, we show in Figures C.2 and C.3 that the forward-imputed search results are closely related. As we discuss in greater detail below, our main results do change somewhat using forward-imputed search results, largely due to differences in aggregated searches in the first month of a presidential administration where the forward- and backward-imputed searches are most different.

search results for the incumbent on a common within-president scale. Overall, searches for the president appear highly variable with high spikes around events like elections and impeachment proceedings. High-profile unilateral actions appear to correlate with small bumps in searches, but these increases are small when viewed against other types of political events.

Findings

We first examine whether salient executive action does, in fact, drive public attention toward the incumbent president. We approach this task with a matched difference-in-differences (DiD) design, which compares periods of regional web-traffic related to the incumbent president that did and did not feature executive action in the news.[14]

"Control" periods represent web-traffic in the same region, occurring at a similar time in a presidential term, and during which no executive action (and accordingly, no related news coverage) took place. To give one concrete example: to assess the effect of coverage of President Trump's executive order on historically black higher education in the South, we would first select a window to examine Google searches for President Trump surrounding the issuance of the action—for instance, 3 days before and 5 days after. Then we would compare the change in search results in the South before and after the action to the change in web-traffic in the same region over another period that did not feature executive action. We detail our procedure for matching windows of time surrounding unilateral actions to control periods in Section D of the SI and show the distributions of the number of matched control periods in Figure D.1. Figure D.2 shows that the matched control periods are largely distributed throughout presidential terms, ensuring we are not comparing to narrow or unique periods of time within presidential terms.

Treatment is equal to whether the given period contained the action (0 or 1) times our measure of media salience (which is scaled to [0, 1]). Consequently, the key independent variable

[14] An alternative approach would be to employ vector auto-regression or error correction models. We avoid this for several reasons. First, our goal is observational causal inference, in which the media shock induced by executive action is the "treatment." VAR and ECMs are not designed to capture this effect. Second, the pooled time-series introduces irrelevant comparisons that make these approaches more sensitive to model specification.

can take values between 0 (no action or a minimally significant action) and 1 (the most significant action). The models include regional and action intercept shifts and we cluster standard errors by action. Due to the reusing of control periods, we also show results in the SI clustered by month.

This research design addresses challenges in analyzing data of this kind. First, it sets aside all differences in web-traffic attributable to the attention-worthiness of presidents themselves or the particular actions they take. The objective, nationwide significance of an action is accounted for by intercept shifts, and the DiD estimate is driven by the salience of that action in the media. Our design also reduces the noise inherent in analyzing web-traffic. As Figure 4 demonstrates, the longer the period, the more likely that some intervening event will swing web traffic wildly toward the incumbent. Moreover, as we move farther away in time from an executive action itself, the measured impact on web-traffic might not be a function of media coverage—but of events precipitated by the action, such as congressional responses or court cases.

Finally, our approach mitigates the potential for bias driven by the strategic timing of executive actions. If, for example, presidents strategically buried their executive actions in already congested news cycles (Djourelova and Durante, 2022), then the baseline search rates will be accounted for by treatment intercept shifts, and the relative change over time would still constitute a valid estimate of the effect of executive action. The only potential cause for concern is if presidents systematically time the issuance of executive action to coincide with some other event that also increases or decreases look-up behavior, such as political scandals.

Note, the design puts several important decisions in the hands of researchers. The two most critical are matching criteria and the length (in days) of comparison windows. Thus, in what follows, we hold constant the former. Cases are matched based on whether they compare the same region, president (Trump or Obama), and time in term (election year or not).[15] For the latter, we report results using 3-day pre-treatment windows and 3, 5, and 7 day post-treatment windows. We opt for asymmetric periods in order to focus attention on the durability

[15]The latter is important, as incumbents might strategically time executive actions for their campaigns (Devdariani and Hirsch, 2023).

of searches for the president while maximizing the number of potential control periods. The shortest length is thought to be the length of a typical news cycle, whereas 7 days is a medium-length window that balances duration with the increasing likelihood of random news events introducing more noise into the analysis.

Our baseline findings, reported in Table 1, demonstrate that media salient executive action does precipitate increased public attention toward the president. The DV is logged to reduce the influence of outlier periods and ease interpretation. Exponentiating the interaction term suggests that the most significant executive actions lead to a 25% increase in web-traffic toward the incumbent, pooling both presidencies.[16] Given the difference-in-differences design, this is an estimate of the average treatment effect on the treated. That is, among the three-day periods immediately following especially significant unilateral actions, Google search interest in the president was on average about 25% higher than it would have been in those same periods had no such action been taken.

One potential concern with a causal interpretation of these results is if the parallel trends assumption is violated. This would occur if, in the absence of an executive order or other type of action being issued, the difference in Google searches across control and treated units does not remain constant. Fully satisfying this assumption requires a long series of pre-treatment search data where unilateral actions were not issued. This is difficult to acquire in our setup due to the frequency of unilateral action issuance by both presidents in our study period. Therefore, we conduct a placebo test of our main findings in Table 1 by comparing searches for the president in the 4-6 days before an action was issued to searches 1-3 days before issuance. The results, shown in Table D.1, suggest that there is no statistically significant increase in Google searches in the days leading up to the issuance of a unilateral action, suggesting that pre trends do not exist. However, the estimates are positively signed and, in the case of the Trump administration, substantively large, possibly due to the frequency of leaks during Trump's first term (Farhi, 2017). Nevertheless, since we find a larger, statistically significant effect during Obama's term despite the lack of large, positive pre-trend bias, we are confident that we are

[16] $e^{0.22} - 1 = 0.25$

Table 1 – Effect of Unilateral Actions on Searches for the President (3 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 3 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are dropped from the analysis. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

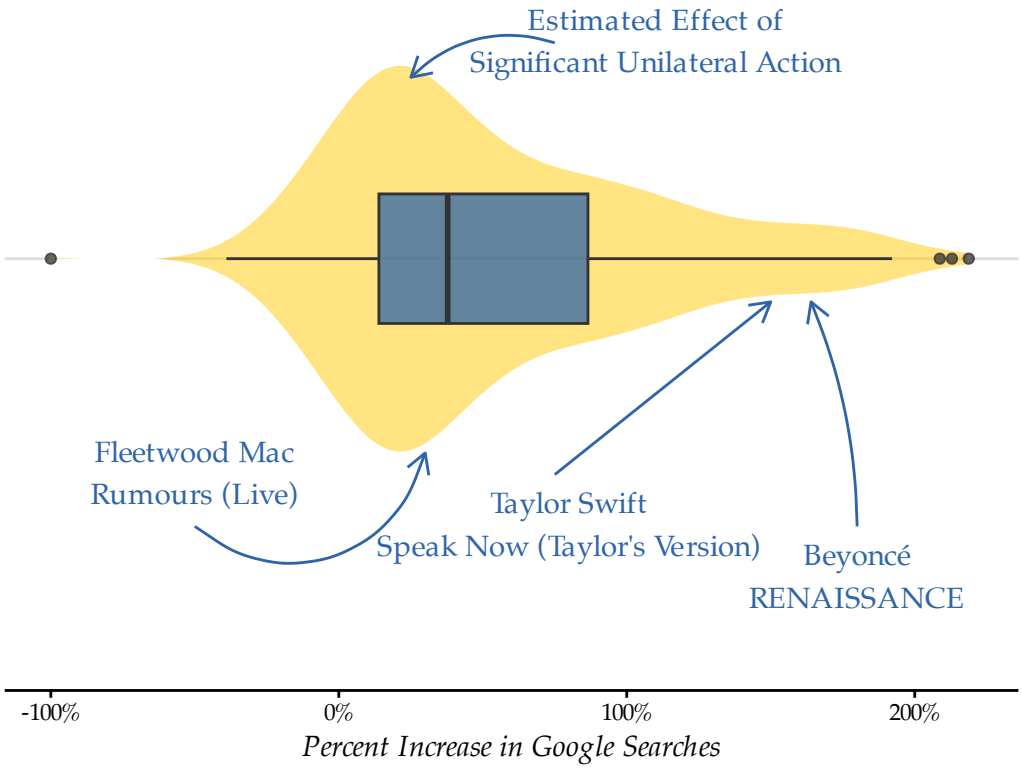
	Pooled	Trump	Obama
Post	-0.40^{***}	-0.37^{***}	-0.40^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.26^{**}	0.37^{**}	0.17
	(0.08)	(0.14)	(0.09)
Post*Treatment	0.22^{***}	0.11	0.28^{***}
	(0.05)	(0.07)	(0.07)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	4122	1917	2205

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

picking up on an underlying effect of actions on Google searches.

Table 1 suggests that searches for the president increase following the issuance of a salient action. To put this effect in context, Figure 5 plots the estimated effect of unilateral action against the percent increase in Google searches following the release of new albums by top-50 pop artists. The 25% increase in searches that follow a significant unilateral action falls between the 25th and 50th percentiles of the distribution. Unilateral actions do not generate the same online attention as new Taylor Swift or Beyoncé albums, but they still attract increased attention commensurate with important cultural events.

Figure 5 – Effect of Significant Unilateral Action Similar to Pop Album Release. Shows the distribution of the percent change in Google searches for top-50 pop artists (as of June 2025) in the 3 days following the release of a new album, relative to the prior 3 days (N = 340). Data on artists and albums retrieved from Spotify's API. Albums issued before January 2009 are dropped, as are instances of multiple albums being issued by the same artist within a 2-week span. Approximately 11% of the albums generated changes in Google searches outside the range of the plot. These values are still included when calculating box-and-whisker summary statistics.



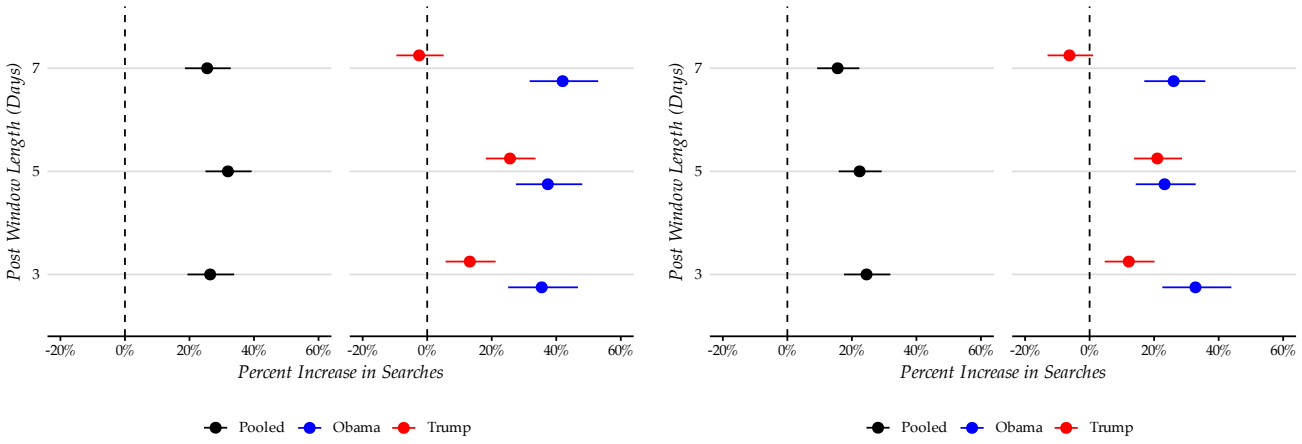
This basic finding is robust to a variety of different modeling approaches. First, a basic limitation of Google Trends data is the censoring of searches with sufficiently low frequencies — ostensibly to protect the anonymity of users. Table 1 treats these units as missing, effectively dropping additional control units from the model. If these cases are treated as zeroes, the effects are similar (SI Table E.5). Second, the results are robust to alternative specifications of the treatment. SI Figure E.2 re-analyzes our data, with the treatment as a dichotomous indicator of whether the action was covered in at least one newspaper article. Again, this effect is positive and statistically significant using 3- and 5-day post-treatment windows. The results also hold

if searches are measured at the national rather than regional level (Table E.1 and Figure E.1) as well as if standard errors are clustered by month instead of action (Table E.4). Finally, the results hold if one uses "President Obama/Trump" rather than "Obama/Trump" as the relevant search terms.

Durability. The next question is whether these effects last. To illustrate this, we report pooled and by-president results for all our intervals, and for both approaches to missing web-traffic periods in Figure 6. The general picture is again that executive action in the news increases web-traffic for the incumbent, and those effects remain for up to 7 days. However, the results vary by president. Regardless of interval length, searches for Obama increase following the release of a unilateral action. In contrast, the effect of unilateral action on searches for President Trump, in addition to often being slightly smaller in magnitude than the estimates for Obama, is only positive in the 3 and 5 days post-action. Over the course of a full week, the percent increase in searches for Trump following unilateral action is indistinguishable from zero.

Different reasons could lie behind this variation across presidents. As reflected in Figure 3, searches for Trump are consistently higher than for other famous figures in sports and entertainment. Thus, while unilateral action might provide Trump with a short-lived increase in Google searches, the size and duration of the effect is limited by the tendency for shorter political news cycles. This may have implications for interpreting our findings out-of-sample, which we return to in the Discussion.

Figure 6 – Significant Actions Lead to Increased Attention. Shows the results of separate regressions testing the interaction between the treatment and post time indicators. The black points show the exponentiated coefficient on the interaction term and corresponding 95% confidence interval for a pooled model that includes a president fixed effect. The red and blue points show the exponentiated interaction coefficients for unilateral actions issued by Trump and Obama, respectively. Full results in Tables 1, E.9, E.7, E.5, E.8, and E.6.



(a) NA-searches assumed zero

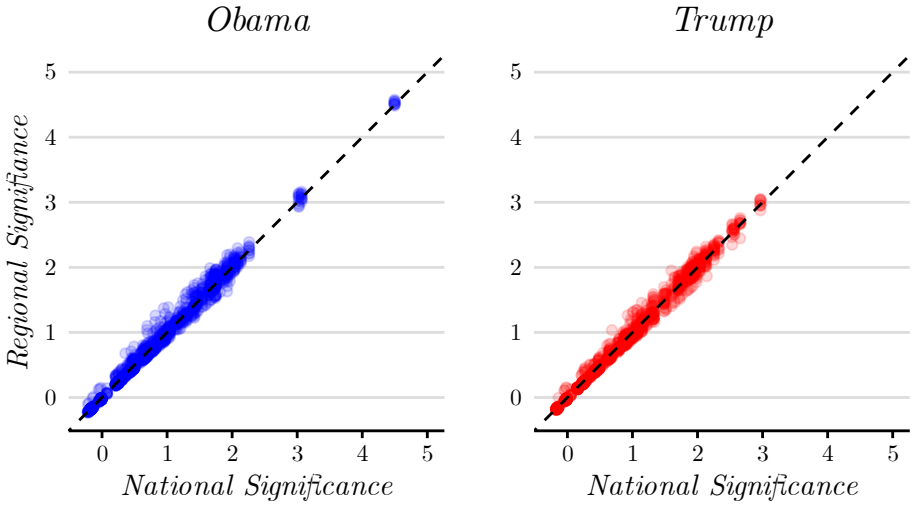
(b) NA-searches dropped

Localism and Nationalization. The previous results set aside all geographic variation in both attention and salience. But this remains an important aspect of the information environment of the public. If the attention spikes we see are driven by regional variation, then in effect, presidents were capturing the attention of the public on the basis of coverage tailored to their area. On the other hand, if minimal regional variation in attention exists, then the president possesses a blunter tool—each executive action would have to be thought of on the basis of its national effect.

To investigate this, we first plot the overall distribution of the posterior means of the regional salience scores z_{jr} against the national salience scores w_j in Figure 7. In general, the correlations between national and regional significance over this period are high. Note, importantly, our hierarchical IRT model does not assume this relationship—it allows these parameters to be independent. What the figure demonstrates is that much of the coverage in regional outlets over this very recent period essentially duplicates what is produced by the large national papers. In some cases, papers with only statewide circulations merely reprint

Associated Press stories about executive action related to the president. The result is an information environment about the incumbent that is *not*, in general, tailored to the particular interests of the region.

Figure 7 – Action salience is nationally driven. Shows the estimates of the posterior means of actions' regional and national salience by president. Actions that did not receive any coverage are dropped.



In summary, we find robust evidence that executive action drives public attention toward the incumbent president, and that this process is mediated by the news. The effects persist beyond the length of an initial news cycle. The evidence is strongest for President Obama, which may be due either to substantive factors unique to him or his presidency. Moreover, the evidence suggests these shifts are typically national.[17] In short, by taking executive action over this period, President Obama and President Trump had measurable impacts on the information available to the American public, nationwide, as well as their attention as measured by search engine traffic.

Discussion

Search Behavior and Presidential Support. We have evidence executive action drives both the salience of policy in the media, and subsequent public attention to the president. But searches

[17] In Section F in the SI we show that there is no evidence of heightened attention to the president across regions.

do not imply public support on their own. As Mellon (2013) shows, the 2012 Republican presidential primary candidate who received the most search traffic was Ron Paul, who did not win a single primary. The important question is under what conditions this incumbent look-up is politically beneficial to the president.

Fortunately, these conditions are explored by the vast theoretical literature on political accountability. In our case, web traffic increases following the issuance of a policy by the incumbent. The decision to investigate something about the incumbent, then, can be thought of as reducing public uncertainty about what is sometimes referred to as the "type", "quality", and "congruence" of the president, relative to the opposition. Models of electoral accountability have shown that the asymmetric reduction of uncertainty is typically enough to advantage the incumbent (e.g., Ashworth, 2012; Judd, 2017). That is, incumbents use their direct sway over policy to ensure the median voter updates their priors in ways favorable to retaining office. Put differently, presidents enact policies they know the public will like, while the opposition is forced to make promises.

This outlines the conditions under which search behavior would *not* translate into a net win for the president. If the president systematically acted in ways that were at odds with the median voter, the attention they brought to themselves would backfire. This could be because they chose bad policies, or because the policies were executed poorly. Alternatively, the look-up behavior might only be a starting place for learning, with internet users visiting sources of information that were systematically biased against the president. Finally, the "public" doing the searching could be non-representative of the median voter. It might be tilted toward the incumbent's current supporters, for example, in which case, the information gained would not translate into an increase in support.

While we have some evidence that cuts against the above conditions, much remains to be studied. Presidents tend to take executive actions which are popular, and the press coverage of their actions tends to be positive (Christenson and Kriner, 2020; Reeves and Rogowski, 2021; Lowande, 2024). Moreover, the idea that searchers would only consume biased information rests on the so-called "echo chamber" hypothesis, which has been mostly refuted by empirical

research on online media consumption (Guess, 2021). However, we do not have individual-level media consumption behavior, which would allow us to take stock of *who* is searching and *what* content they ultimately consume. Both factors are important for determining whether the boost in attention translates to political support.

Generalizability. One natural question is whether these findings would be different in earlier presidencies, or for the most recent Biden and Trump administrations. Of course, any extrapolation beyond these historical periods and political actors carries risks. However, there are several reasonable expectations in this case. Differences in effect sizes between Obama and Trump might represent a trend. That is, earlier presidencies might show greater impacts of executive action on public attention. Over time, as executive actions become more frequent, they may be more routine in the public mind, and therefore, less likely to draw attention. A related point is that the second Trump administration has so thoroughly dominated the news cycle, and executive action has occurred so frequently, that it is possible there would essentially be no control periods. We would have to compare more or less salient actions to each other, and have difficulty recovering the appropriate counterfactual.

Our study also highlights fruitful ground for additional work on public "up-take" of policy information. Because we study public attention at the regional level, we cannot speak directly to factors like partisanship, ideology, and political knowledge, which might contribute directly to the consumption of news about executive action. In other words, we do not know what segment of the president's constituency is driving these shifts in web attention. As we have noted, these questions are critical for understanding whether the advantage in attention translates to an advantage in acquiring political support.

Conclusion

Scholars have long acknowledged that democracies function well or poorly on the basis of available information. Voters need information to judge the performance of incumbents and hold them accountable. But separation of powers systems fracture the source of policy. Incum-

bents are often in a contest to take credit for government action that benefits constituents, and lay blame for things that do not. Our study has made a simple argument: Chief executives possess tools of the state that allow them to uniquely influence what information is available to voters. That information subsequently centers the political attention of the public on them. Executive action issued during the presidency of Barack Obama and the first term of President Trump increased subsequent search engine traffic by 25%, on average.

More generally, our findings suggest scholarship on democracy and political communication could benefit from attention to institutions and policy outputs. Presidents' agenda setting power might have implications for their basic accountability relationship between the state and voters. This also leads naturally to questions about whether decree authority offers similar benefits in other democracies, and in sub-national settings. American state governors now routinely issue executive actions, and the study of executive action itself began with a focus on Latin American democracies. Both settings offer institutional variation which is absent in the American presidency, and might yield more knowledge about the conditions under which presidential power influences the information available in democratic societies.

References

Ashworth, Scott. 2012. "Electoral Accountability: Recent Theoretical and Empirical Work." *Annual Review of Political Science* 15:183–201.

Bleher, Johannes and Thomas Dimpfl. 2022. "Knitting Multi-Annual High-Frequency Google Trends to Predict Inflation and Consumption." *Econometrics and Statistics* 24:1–26.

Bolton, Alexander and Sharece Thrower. 2022. *Checks in the Balance: Legislative Capacity and the Dynamics of Executive Power*. Princeton University Press.

Boydstun, Amber E. 2013. *Making the News: Politics, the Media, and Agenda Setting*. University of Chicago Press.

Bullock, John, Alan Gerber, Seth Hill and Gregory Huber. 2015. "Partisan Bias in Factual Beliefs about Politics." *Quarterly Journal of Political Science* 10(4):519–578.

Campbell, Angus, Philip E. Converse, Warren E. Miller and Donald E. Stokes. 1960. *The American Voter*. University of Chicago Press.

Canes-Wrone, Brandice. 2006. *Who Leads Whom? Presidents, Policy, and the Public*. University of Chicago Press.

Carey, John M. and Matthew Soberg Shugart. 1998. *Executive Decree Authority*. Cambridge University Press.

Caughey, Devin and Christopher Warshaw. 2022. *Dynamic Democracy: Public Opinion, Elections, and Policymaking in the American States*. University of Chicago Press.

Chiou, Fang-Yi and Lawrence S. Rothenberg. 2014. "The Elusive Search for Presidential Power." *American Journal of Political Science* 58(3):653–668.

Chiou, Fang-Yi and Lawrence S. Rothenberg. 2017. *The Enigma of Presidential Power: Parties, Policies and Strategic Uses of Unilateral Action*. Cambridge University Press.

Christenson, Dino P. and Douglas L. Kriner. 2020. *The Myth of the Imperial Presidency: How Public Opinion Checks the Unilateral Executive*. University of Chicago Press.

Cooper, Phillip J. 2014. *By Order of the President: The Use and Abuse of Executive Direct Action*. 2 ed. University Press of Kansas.

Curry, James, Frances Lee and Robert Oldham. 2024. "On the Congress Beat: How the Structure of News Shapes Coverage of Congressional Action." *Political Science Quarterly* .

Dancy, Geoff and Christopher Fariss. 2023. "The Global Resonance of Human Rights: What Google Trends Can Tell Us." *American Political Science Review* 118(1):252–273.

Deans, Bob. 1997. "IN CENTRAL AMERICA, SONGBIRDS REPLACE CIVIL WAR: [SOONER Edition]." *Pittsburgh Post - Gazette* p. 4.

Devdariani, Saba and Alexander V. Hirsch. 2023. "Voter Attention and Electoral Accountability." *Journal of Public Economics* 224:104918.

Djourelova, Melena and Ruben Durante. 2022. "Media Attention and Strategic Timing in Politics: Evidence from U.S. Presidential Executive Orders." *American Journal of Political Science* 66:813–834.

Dresser, Michael. 2011. "Red Line Picked for Faster Review: East-west Rail among 14 Projects Chosen Nationwide." *The Baltimore Sun* p. 6.

Edwards, George C. 2006. *On Deaf Ears: The Limits of the Bully Pulpit*. Yale University Press.

Eichenauer, Vera Z., Ronald Indergand, Isabel Z. Martínez and Christoph Sax. 2021. "Obtaining Consistent Time Series from Google Trends." *Economic Inquiry* 60(2):694–705.

Farhi, Paul. 2017. "The Trump administration has sprung a leak. Many of them, in fact." *The Washington Post* .

Franco, Annie, Justin Grimmer and Chloe Lim. 2018. "The Limited Effect of Presidential Public Appeals." *Working paper, Stanford University* pp. 1–38.

Garz, Marcel and Gregory J. Martin. 2021. "Media Influence on Vote Choices: Unemployment News and Incumbents' Electoral Prospects." *American Journal of Political Science* 65(2):278–293.

Gelman, Andrew and Donald B. Rubin. 1992. "Inference from Iterative Simulation Using Multiple Sequences." *Statistical Science* 7(4):457–472.

Gentzkow, Matthew, Jesse M. Shapiro and Michael Sinkinson. 2011. "The Effect of Newspaper Entry and Exit on Electoral Politics." *American Economic Review* 101(7):2980–3018.

Gentzkow, Matthew, Jesse M Shapiro and Michael Sinkinson. 2014. "United States Newspaper Panel, 1869-2004." *Inter-university Consortium for Political and Social Research [distributor]* .

Ginsberg, Jeremy, Matthew H. Mohebbi and Rajan S. Patel. 2009. "Detecting Influenza Epidemics Using Search Engine Query Data." *Nature* 457:1012–1014.

Gramlich, John and Kirsten Eddy. 2024. "The Link Between Local News Coverage and Americans' Perceptions of Crime." Pew Research Center. **URL:** *https://www.pewresearch.org/short-reads/2024/08/29/the-link-between-local-news-coverage-and-americans-perceptions-of-crime/*

Grimmer, Justin and B. M. Stewart. 2013. "Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts." *Political Analysis* 21(3):267–297.

Grimmer, Justin, Sean J Westwood and Solomon Messing. 2013. *The Impression of Influence : How Legislator Communication and Government Spending Cultivate a Personal Vote*. Princeton University Press.

Guess, Andrew. 2021. "(Almost) Everything in Moderation: New evidence on Americans' online media diets." *American Journal of Political Science* 65:107–1022.

Hopkins, Daniel J. 2018. *The Increasingly United States: How and Why American Political Behavior Nationalized*. University of Chicago Press.

Howell, William G. 2003. *Power without Persausion: The Politics of Direct Presidential Action*. Princeton University Press.

Howell, William G., Ethan Porter and Thomas J. Wood. 2020. "Rethinking Public Appeals: Experimental Evidence on Presidential Performances." *Journal of Political Institutions and Political Economy* 1(1):137–158.

Iyengar, Shanto, Yphtach Lelkes, Matthew Levendusky, Neil Malhotra and and Sean J. Westwood. 2019. "The Origins and Consequences of Affective Polarization in the United States." *Annual Review of Political Science* 22:129–146.

Judd, Gleason. 2017. "Showing off: Promise and Peril in Unilateral Policymaking." *Quarterly Journal of Political Science* 12(2):241–268.

Kernell, Samuel. 2007. *Going Public: New Strategies of Presidential Leadership*. SAGE Publications.

Kinder, Donald R. and Nathan P. Kalmoe. 2017. *Neither Liberal nor Conservative: Ideological Innocence in the American Public*. University of Chicago Press.

Lowande, Kenneth. 2024. *False Front: The Failed Promise of Presidential Power in a Polarized Age*. University of Chicago Press.

Lowande, Kenneth and Jon C. Rogowski. 2021. "Presidential Unilateral Power." *Annual Review of Political Science* pp. 1–25.

Lowande, Kenneth and Jon C. Rogowski. 2022. "Executive Power in Crisis." *American Political Science Review* .

Major, Mark. 2014. *The Unilateral Presidency and the News Media: The Politics of Framing Executive Power*. The Evolving American Presidency New York: Palgrave Macmillan.

Mayer, Kenneth R. and Kevin Price. 2002. "Unilateral Presidential Powers: Significant Executive Orders, 1949-99." *Presidential Studies Quarterly* 32(2):367–386.

Mayhew, David R. 1974. *Congress: The Electoral Connection*. Yale University Press.

McCombs, Maxwell E. and Donald L. Shaw. 1972. "The Agenda-Setting Function of Mass Media." *Public Opinion Quarterly* 36:176–187.

Mellon, Jonathan. 2013. "Where and When CanWe Use Google Trends to Measure Issue Salience?" *PS: Political Science and Politics* pp. 280–290.

Moe, Terry M. and William G. Howell. 1999. "The Presidential Power of Unilateral Action." *Journal of Law, Economics, and Organization* 15(1):132–179.

Nakamura, David. 2013. "Outside Beltway, Obama Seeks a Reset." *The Washington Post* p. 2.

Noble, Benjamin G. 2024. "Presidential Negative Partisanship." *Working Paper, UCSD* pp. 1–42.

O'Keefe, Ed. 2010. "Changes in Leave Policies Go to Opposite-Sex Partners; Stepparents Also among Those Eligible for Illness, Funeral Benefits." *The Washington Post* p. 3.

Peterson, Erik. 2021. "Paper Cuts: How Reporting Resources Affect Political News Coverage." *American Journal of Political Science* .

Press, The Associated. 2000. "SUPREME COURT HEARS ARGUMENTS OVER GOVERNMENT'S STANDARDS FOR CLEAN AIR JUSTICES APPEAR SKEPTICAL ABOUT WHETHER EPA MUST CONSIDER COSTS TO INDUSTRY: [FIVE STAR LIFT Edition]." *St. Louis Post - Dispatch* p. 2.

Priest, Dana and Walter Pincus. 2004. "Quick Fixes on Intelligence Considered; Administration Explores Short-Term Remedies to Diffuse Political Pressure From 9/11 Report: [FINAL Edition]." *The Washington Post* p. 05.

Prior, Markus. 2007. "The Real Media Divide." *Washington post (Washington, D.C. : 1974)* .

Reeves, Andrew and Jon C. Rogowski. 2021. *No Blank Check: Why the Public Dislikes Presidential Power and What It Means for Governing*. Cambridge University Press.

Rudalevige, Andrew. 2021. *By Executive Order: Bureaucratic Politics, Presidential Management, and the Limits of Unilateralism*. Princeton University Press.

Soroka, Stuart and Christopher Wlezien. 2010. *Degrees of Democracy*. Cambridge University Press.

Soroka, Stuart and Christopher Wlezien. 2024. "Media Reflect! Policy, the Public, and the News." *American Political Science Review* 118:1563–1569.

Soroka, Stuart N. and Christopher Wlezien. 2022. *Information and Democracy: Public Policy in the News*. Cambridge University Press.

Stephens-Davidowitz, Seth. 2013. "The Cost of Racial Animus on a Black Presidential Candidate: Using Google Search Data to Find What Surveys Miss." *Working Paper, Harvard University* .

Tilove, Jonathan. 2015. "Mammoth Site Designated a New National Monument: Dig along Bosque River Features a Variety of Fossils." *Austin American Statesman* p. 3.

Tulis, Jeffery. 1987. *The Rhetorical Presidency*. Princeton University Press.

Zhao, Mengtian, Heather Rosoff and Richard S. John. 2019. "Media Disaster Reporting Effects on Public Risk Perception and Response to Escalating Tornado Warnings: A Natural Experiment." *Risk Analysis* 39(3):535–552.

Supplementary Information
Learning About Presidential Power

A Collecting Newspaper Coverage Data **SI-1**
A.1 Procedure for collecting coverage SI-1
A.2 Procedure for coding coverage SI-1
A.3 Compiling Coverage Data SI-4

B Salience Model Diagnostics **SI-6**

C Google Trends **SI-8**

D Analysis Diagnostics **SI-11**

E Additional Results **SI-16**

F Regional Partisanship Results **SI-25**

A Collecting Newspaper Coverage Data

A.1 Procedure for collecting coverage

Our primary goal was to generate an index of news coverage of unilateral action that would be broadly representative of print journalism during this period. We first generated a list of the most circulated papers in each state and the District of Columbia using panel data collected by (Gentzkow, Shapiro and Sinkinson, 2011). We then found the paper with the largest circulation in each state that was included in ProQuest's news database. Finally, we added any newspapers in the top 25 publications by national circulation not already included, regardless of state. The result of this inclusion procedure is the outlets listed in Table A.1. Our main concern is that our selection procedure would over-sample newspapers from large cities, and thus, lean Democratic in its endorsements and affiliations. This might lead to coverage differences across presidents attributable to the uniqueness of the sample. However, only 14 of the 51 newspapers have Democratic partisan affiliations (Gentzkow, Shapiro and Sinkinson, 2011).

Our sample does not include a newspaper from 8 states: Idaho, Utah, Alaska, North Dakota, Oklahoma, Arkansas, North Carolina, and Rhode Island. There are two dynamics at play. First, some states' major paper is simply not part of the ProQuest database. Including them would be prohibitively time consuming. This is most common in states where the largest paper in terms of circulation is relatively small in terms of its state readership proportion. Second, during this period, the Washington Post and New York Times had subscriber bases in adjacent states that rivaled or exceeded the major state newspaper. Though we do not investigate this in this study, our time series allow us to examine secular changes in presidential news coverage due, in part, to the nationalization of reporting and the financial success of these major outlets.

A.2 Procedure for coding coverage

Prior to being given complete coding sheets, our researcher assistants attended a 1 hour training and then were assigned to complete a short coding sheet of actions which took roughly 4 hours, on average. After that, they were provided a set of correct coverage answers, and were tasked with going through each action to reconcile differences in their coding and the correct sheet. Any remaining discrepancies were then reviewed in a meeting with the principal investigators. In addition to this initial screening, we monitored inter-coder reliability in real-time, as the coders completed actions. If any significant discrepancies emerged, we examined their work and then met with the research assistant to clarify our coding rules.

We provided the following instructions to undergraduate research assistants:

"The csv contains a list of executive actions. We want to know how the media covered these cases. You will use the ProQuest database to pull newspaper coverage. Read these instructions in their entirety before proceeding. If you have any questions, contact (blinded for peer review).

What are you looking for?

We are looking for U.S. news coverage (excluding opinion and reprint pieces) of executive action. What does this mean? U.S. presidents routinely sign documents that we call "directives" (like executive orders, memoranda, proclamations, etc.) that are formal actions that do not require approval from Congress. These directives do all kinds of things, like build border walls, establish commissions, raise tariffs, and create national monuments. We want to find the news stories that cover or mention these executive actions. These are distinct from news

stories that mention legislation in Congress, speeches, meetings with international leaders, or the president's generic position on some topic.

Examples of what counts as relevant coverage:

The action is the main subject of the story. Suppose you were to collect articles about the unilateral action titled "Memorandum on Extension of Benefits to Same-Sex Domestic Partners of Federal Employees." In O'Keefe (2010) in the Washington Post, the action is the main subject of the story. The explicit attribution of the memorandum comes in the second paragraph: "The policy change, published in Monday's Federal Register, is part of reforms ordered last year by President Obama when he extended fringe benefits to the same-sex partners of gay federal workers. . . " Note, it does not explicitly say "memorandum" – but we know because of the timing and content that this is what the news story is referring to.

The action is not the main subject, but appears in the text somewhere after the lede. The document itself is mentioned. Suppose you were to collect articles about the unilateral action titled "Memorandum on Speeding Infrastructure Development Through More Efficient and Effective Permitting." In Dresser (2011) in The Baltimore Sun, the memo is not the main subject, but it is mentioned explicitly toward the end of the article: "The announcement follows a memorandum President Barack Obama signed in August directing federal agencies to expedite environmental reviews and permit decisions for projects..."

The action is not the main subject, but appears in the text somewhere after the lede. The document itself is not mentioned. Suppose you were to collect articles about the unilateral action titled "Proclamation 9298-Establishment of the Berryessa Snow Mountain National Monument." In Tilove (2015) in the Austin-American Statesmen, Obama's designation of Berryessa Snow Mountain is mentioned, even though the article is about the Waco Mammoth Site. It reads: "It was one of three new national monuments created Friday by the president. The other two are Berryessa Snow Mountain in California, a landscape containing..." Notice, the article does not mention a proclamation specifically, but it counts as coverage because the thing the proclamation did – create the specific national monument – is mentioned.

The action is not the main subject, but appears in the text somewhere after the lede. The document itself is not mentioned because the story appears before it was signed. Suppose you were to collect articles about an executive order that made the head of the CIA more prominent in the NSC. In Priest and Pincus (2004) in the Washington Post, the executive order and what it does is referenced vaguely because it had not yet been signed. The article reads: "President Bush, at his ranch in Crawford, Tex., held another video conference yesterday with his national security advisers to discuss a set of executive orders he plans to issue next week..." If you think you have a case like this, you can double check this by looking at the date of the directive and the date of the article.

Examples of what does not count as relevant coverage:

The article references a different directive on the same topic. Suppose you were to collect coverage of the unilateral action titled "Memorandum on Speeding Infrastructure Development Through More Efficient and Effective Permitting." In Nakamura (2013) in the Washington Post, a different memo on a similar topic is mentioned: "In Baltimore, he announced that he had signed a memorandum to speed up permits for infrastructure projects, which he said would help get more unemployed workers back on the job." The telltale sign is that this language

mentions a memo signed within the last week, but this article was published two years after the memorandum you're looking for coverage of. This is another example of how the timing of the action can be a guide for whether the coverage is relevant.

The article references the same topic, but not the action the president took. Suppose you were to collect coverage of the Presidential Review Directive titled "U.S. Environmental policy in Latin America and the Caribbean." In Deans (1997) in the Pittsburgh Post-Gazette, the President's meetings, policies and positions on the environment and Latin America are mentioned, but this directive is not. There is no mention of an order, or a review directive. This is not relevant coverage because it is not a news story about the president's specific action in the Presidential Review Directive.

The article references the action but does not attribute it to the President or his administration. Suppose you were to collect coverage of the memorandum titled "Memorandum on Implementation of Revised Air Quality Standard for Ozone and Particulate Matter." The memo asks the EPA to implement stricter particular matter rules. Press (2000) mentions that a lower court overturned the EPA's stricter particular matter rules, but it attributes this policy failure to the EPA, not the president or his administration. This is not relevant because it does not connect the failure to the unilateral action taken by Clinton.

Search Procedures:

Each row in the csv file pertains to a different presidential action. The first 13 variables are already filled in and contain information to help you identify the action and search for it in ProQuest. Do not replace or alter any of this information. You are going to use the advanced search function to see if any newspaper articles cover the relevant action.

1. Access the ProQuest database using your credentials.
2. If it is not already, sort the spreadsheet by date (year, month, day), this will make adjusting the publication dates for the advanced searches easier.
3. Adapt the advance search for coverage of particular action:
    - Publication date: Specific date range (three months before the action was issued) to (last day president in office, January 20 XXXX)
    - Main search line: Copy and paste the search syntax provided. If this syntax isn't exact, the search will be wrong.
    - Limit to: Full text
    - Sort results by: "Relevance"
    - Exclude duplicate documents.
4. Input the total number of results in the total search results field, which is named 'no.results' in the spreadsheet.
5. Select "Get Search Link" and copy/paste this link in the search link field, which is named 'search.link' in the spreadsheet.
6. Place check marks on all relevant articles.
    - First look at the one sentence summary of the action, as well as the directive type. These will guide you as you examine the coverage returned by the advanced search.
    - To identify relevant articles:
        - . Click on each entry and examine the full text.
        - Control-F the words "president," "administration," and the president's name (e.g., Obama).
        - An entry is relevant if it mentions the directive. Note:

        * See the examples. It may or may not explicitly mention the document. But it must mention the president or his administration and the specific action.
        * Articles published outside the US are not relevant. The Times of London (a UK publication) and the Gazette (published in Montreal) will occasionally appear in search results.
    - Opinion pieces or letters to the editor are not relevant. Do not check them. (Sometimes you can only tell if an article is an opinion piece by reading the introduction or conclusion for any supportive/unsupportive language.)
    - In the case of articles published before the issuance of the action, look to see if the article mentions the upcoming order or discussions about the order. (e.g. "A source in the Bush administration says that the president will impose steel tariffs next week.")
    - Transcripts of speeches are not relevant.
    - Other helpful tips:
        - If the first 10 entries do not reference the action, it is very unlikely any others will.
        - Documents with the "Federation of American Scientists" source are very unlikely to have received coverage. These documents are generally top secret upon issuance, and news outlets only learn of their existence long after the fact.
        - To limit number of search results:
            * Check all duplicate publications with the same title, which are usually local news articles reprinting national news providers.
            * Do not check duplicate entries within the same news source.
            * Do not check REPRINTING coverage. These are entries that simply reprint the government document in full, without commentary.
7. If there is no relevant coverage, enter 0 in the coverage field and move on to the next document, otherwise, enter a 1.
8. Export the results to an .xls file.
    - Click "XX selected articles" link at the top of the page.
    - Select "All save options" in the upper right corner.
    - Select "Download XLS"
    - Save the file as a .csv in the "coverage/lastname" folder with the following naming convention: "uid.csv"
9. Uncheck all checked articles before proceeding to the next case."

We randomly selected roughly 1 of every 3 actions to be coded be two coders. In general, agreement and reliability for whether there was coverage of a particular action was very high, with 94% agreement and a Krippendorf's alpha of .87.

A.3 Compiling Coverage Data

We compiled the coverage data iteratively in order to check our coders' work and catch mistakes early. For actions that were double-coded, we only included them in our final dataset if the coders were in agreement that the action did or did not receive coverage. This resulted in dropping 64 actions from our final dataset. Second, if two coders agreed that coverage existed for an action but disagreed on the number of articles providing coverage, we set the amount of coverage equal to the larger of the two numbers. We opted to use the larger of the two values since it is conservative relative to our main argument.

Table A.1 – Our Sample of Newspapers is Geographically and Politically Diverse Lists the newspapers in our sample along with their Democratic (D), Republican (R), or Independent (I) partisan affiliations, as provided by Gentzkow et al (2011). Papers are classified as Republican (Democrat) if they ever declared an affiliation with the party.

State	City	Publication	Partisan Affiliation
AL	Montgomery	Montgomery Advertiser	D
AZ	Phoenix	Arizona Republic	R
CA	Los Angeles	Los Angeles Times	R
CA	San Jose	San Jose Mercury News	NA
CO	Colorado Springs	The Gazette	R
CT	Hartford	Hartford Courant	R
DC	Washington	The Washington Post	I
DE	Wilmington	The News Journal	I
FL	Orlando	Orlando Sentinel	I
FL	Tampa Bay	Tampa Bay Times	D
GA	Atlanta	The Atlanta Journal Constitution	NA
HI	Honolulu	Honolulu Advertiser	R
IA	Des Moines	Des Moines Register	R
IL	Chicago	Chicago Tribune	R
IN	Indianapolis	Indianapolis Star	R
KS	Topeka	Topeka Capital Journal	NA
KY	Louisville	Courier Journal	D
LA	Shreveport	The Times	D
MA	Boston	Boston Globe	I
MD	Baltimore	The Baltimore Sun	I
ME	Portland	Portland Press Herald	R
MI	Detroit	Detroit Free Press	D
MI	Detroit	Detroit News	R
MN	Minneapolis	Star Tribune	I
MO	St. Louis	St Louis Post Dispatch	D
MS	Jackson	The Clarion Ledger	D
MT	Billings	The Billings Gazette	R
NE	Lincoln	Lincoln Journal Star	I
NH	Manchester	The Union Leader	R
NJ	Bergen County	The Record	R
NM	Albuquerque	Albuquerque Journal	I
NV	Las Vegas	Las Vegas Review Journal	D
NY	New York	New York Daily News	I
NY	New York	New York Times	D
NY	New York	Wall Street Journal	NA
OH	Cincinnati	Cincinnati Enquirer	D
OR	Salem	Statesman Journal	I
PA	Philadelphia	Philadelphia Inquirer	R
PA	Pittsburgh	Pittsburgh Post Gazette	R
SC	Charleston	The Post And Courier	I
SD	Sioux Falls	Argus Leader	R
TN	Nashville	The Tennessean	D
TX	Austin	Austin American Statesman	D
VA	Newport News	Daily Press	D
VT	Burlington	The Burlington Free Press	R
WA	Spokane	Spokesman Review	R
WI	Milwaukee	Milwaukee Courier	NA
WI	Milwaukee	Milwaukee Journal Sentinel	I
WV	Charleston	Charleston Gazette Mail	NA
WV	Charleston	The Charleston Gazette	D
WY	Cheyenne	Wyoming Tribune Eagle	NA

B Salience Model Diagnostics

Figure B.1 – $\hat{R}$ Values Show High Degree of Convergence. Shows the distributions of the potential scale reduction factors for the w, omega^2, beta, and alpha parameters.

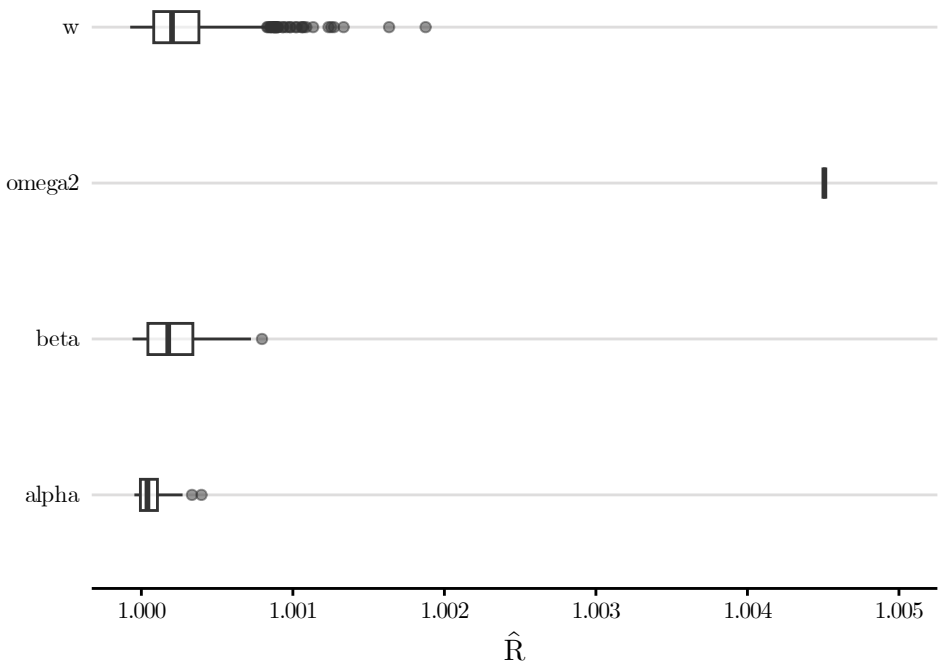


Figure B.2 – Difficulty Parameters. Plots the difficulty parameters from the IRT model.

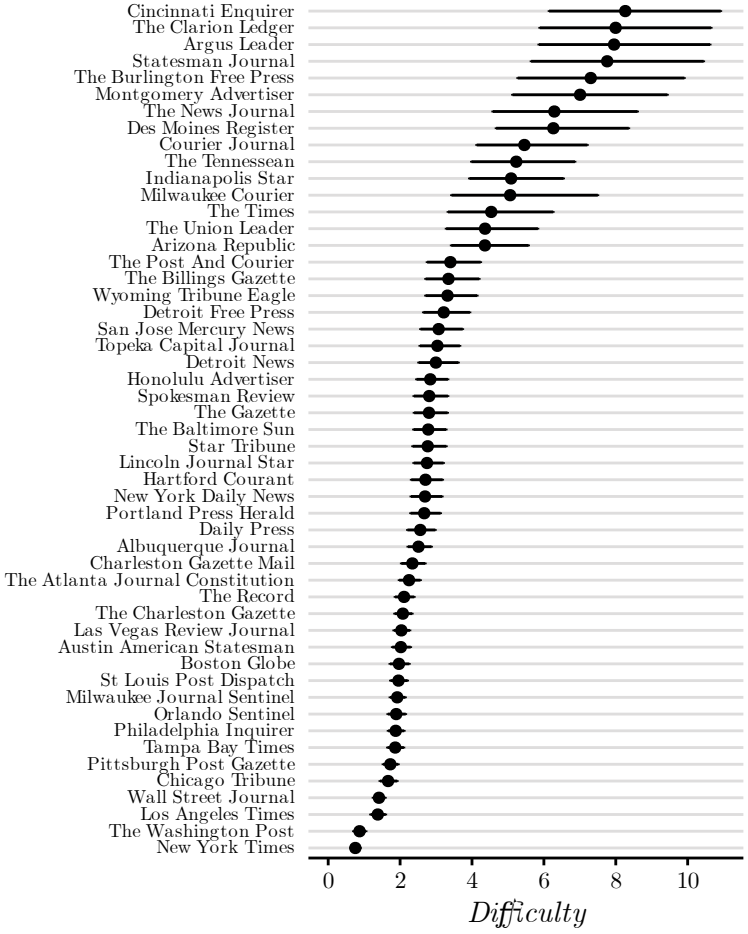
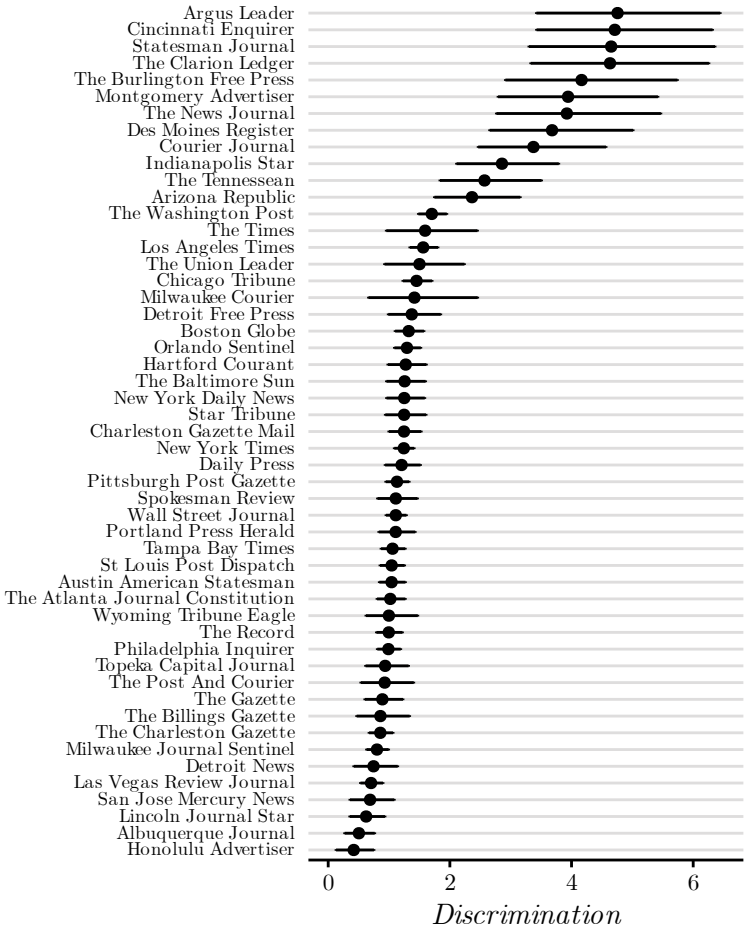


Figure B.3 – Discrimination Parameters. Plots the discrimination parameters from the IRT model.



C Google Trends

We estimate regional and national day-level search results for "Obama" and "Trump" using the following steps:

- Using SerpAPI, download Google Trends search results for "Obama" and "Trump" at daily-, weekly-, and monthly-intervals in each state and nationwide. The data are downloaded in overlapping intervals:
  1. Daily data are downloaded in 6-month chunks. Adjacent chunks have 5 months of overlapping data. For instance, the first chunk of Obama searches include January 20, 2009 through July 20, 2009. The next chunk covers February 20, 2009 through August 20, 2009.
  2. Weekly data are downloaded in 24-month chunks, except when they go up to the end of a president's term in which case they encompass a longer period. Adjacent chunks have 12 months of overlapping data. For instance, the first chunk of weekly searches for Trump cover January 27, 2017 through January 20, 2019 and the second chunk encompasses January 20, 2018 through January 19, 2021.
  3. Monthly data can be extracted on a consistent scale for any time period over 5 years.

Therefore, it is only available for Obama.

- For state-level data, aggregate searches up to the regional level (See Figure 1). For the states within each region, calculate the mean number of searches for the time period weighted by the state's population size.
- Truncate data to exclude periods of consistent zero hits as this effectively breaks the imputation algorithm. Keep all weekly data up to an instance of three straight weeks of zero searches. Keep all daily data up to an instance of 60 straight days of zero search results. If a region has less than 40 days or weeks with any searches it is completely dropped.
- Run the imputation algorithm defined in the main text in Equation 4 within each geographical region or nationwide.
- Sum up the number of imputed daily hits within the 3-, 5-, or 7-day interval defined before/after unilateral actions and their control dates.

Figure C.1 – Searches for "Obama (Trump)" versus "President Obama (Trump)". Shows the raw search results for "Obama" versus "President Obama" and "Trump" versus "President Trump". Searches for a given president are on a common 0-100 scale. Results for Obama are at monthly intervals, while results for Trump are at weekly intervals.

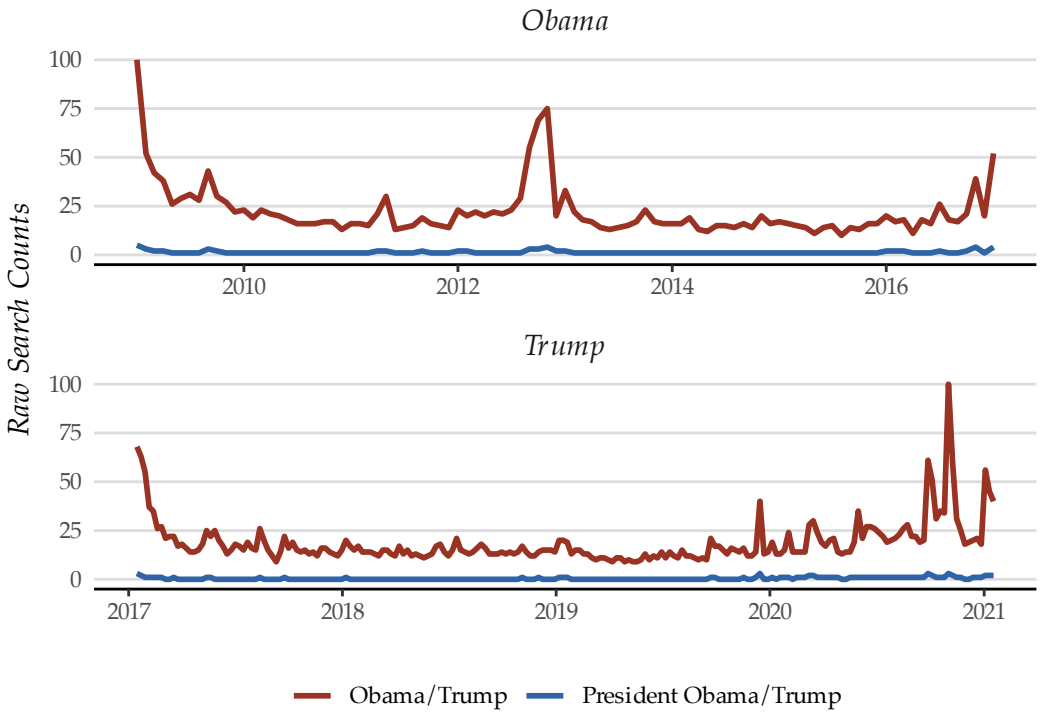


Figure C.2 – Google Searches for Presidents Obama and Trump (Alternative Algorithm). Shows Google searches for "Obama" and "Trump" during their respective terms on a common scale. Search counts cannot be compared across presidents. Counts are national and calculated using the smoothing algorithm run from the earliest to the most recent period.

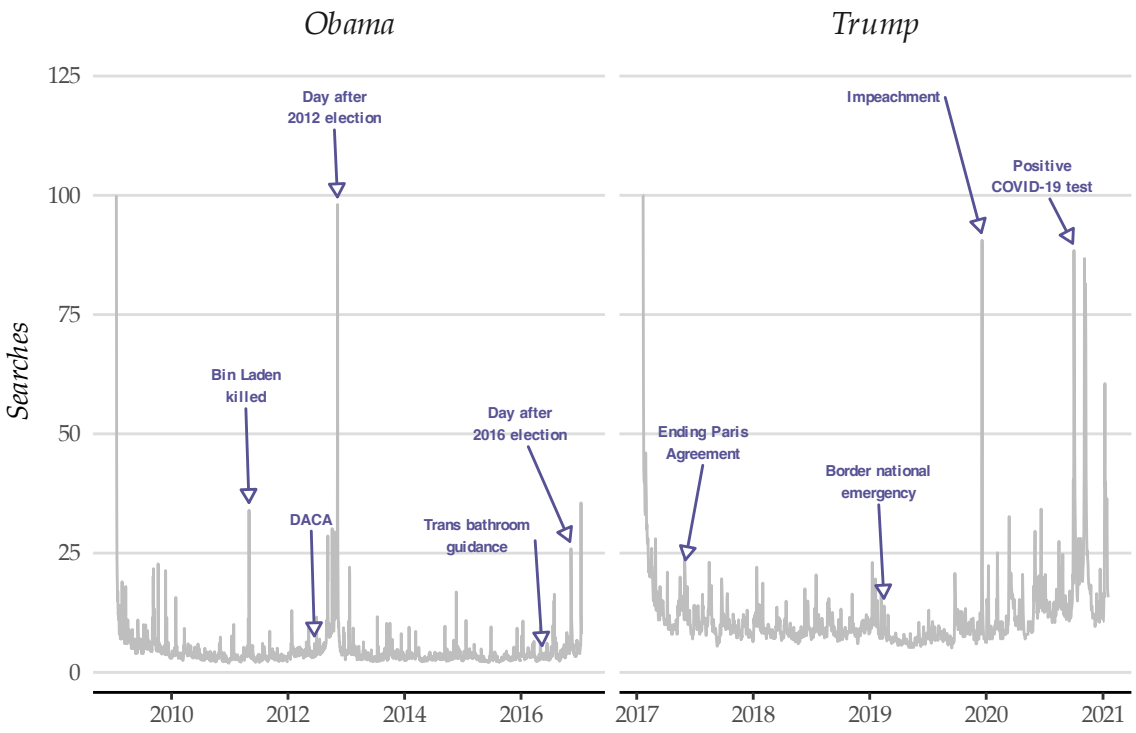
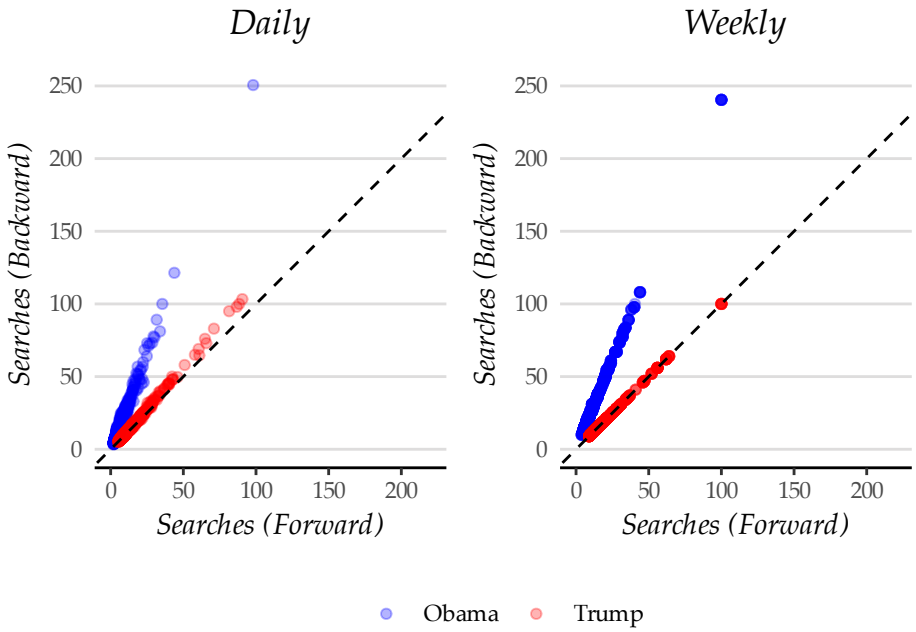


Figure C.3 – Correlation Between Imputation Algorithms. Plots the daily and weekly search results using the algorithm run from the most recent period to the earliest (backward) and earliest period to most recent (forward). The correlation between the daily search results is 1.32 (R^2 = 0.98) and the correlation between the weekly search results is 1.08 (R^2 = 0.92).



D Analysis Diagnostics

We match control periods to unilateral actions using the following algorithm. For simplicity, we walk through an example of finding all control periods for President Obama's DACA order, which was issued on 6-15-2012. The pre/post periods are set to 3 and i days, respectively (where i is either 3, 5, or 7 days).

1. Define the set of possible control periods by first finding all dates from 6-15-2012 to 1-20-2017 at $i+3$ day intervals and 6-15-2012 to 1-20-2009 at $-(i+3)$ day intervals. The unique concatenation of these dates form the beginning of each control period. The center of the potential control periods equals the beginning dates plus 3 days and the end boundary dates equal the beginning dates plus $i+3-1$ days. For instance, one possible control period is comprised of the beginning, center, and end dates of 10-21-2012, 10-24-2012, and 10-28-2012.
2. Define the beginning, center, and end dates of all unilateral actions issued by the given

president. The center date is the date each action was issued. The beginning date is 3 days prior and the end date is $i - 1$ days after.

3. Using the set of possible control periods and the set of action periods, find all control periods that do not overlap with any action periods. In R, this is done with the chunk of code below, where `control_intervals` is a dataframe containing the pre, center, and post dates of each potential control period; `action_intervals` is a dataframe containing the pre, center, and post dates of each unilateral action issued by Obama; and `control_pre`, `control_post`, and `action_post` are the pre/post boundary date variables in, respectively, `control_intervals` and `action_intervals`.

```
anti_join(control_intervals,
          action_intervals,
          join_by(overlaps(control_pre,
                           control_post,
                           action_pre,
                           action_post,
                           bounds = "[]")))
```

4. Since DACA was issued in a re-election year, remove any remaining control periods with central dates not in 2012.[18] The remaining control periods comprise the set of matched control periods for DACA using a post-treatment window of i days.
5. Sum the number of day-level Google Trends searches within the pre/post interval for the treatment and control periods. The pre interval includes all days up to but not including the central date. The post interval includes the central date up to and including the post boundary date.

This process is then repeated for all other unilateral actions in our dataset. The following figures show the distribution of the number of matched control periods by region, nationally, and nationally by president. The horizontal axis denotes the number of control periods a particular action was matched to, while the vertical axis indicates the frequency (i.e., the number of unilateral actions matched to x number of control periods). Figures D.3 and D.4 show the share of unilateral actions in our dataset successfully matched to at least one control period over time.

[18] Actions issued in 2020 receive a similar treatment. Actions issued in Obama's first (second) term are also only matched to control periods in his first (second) term.

Figure D.1 – Number of Matched Control Units, by President. Plots the number of control units matched to unilateral actions using post-treatment windows of 3, 5, and 7 days. The pre-treatment interval is 3 days. Control units computed using national-level Google Trends search results. Results are the same using region-level search results. The bin width is set to 2.

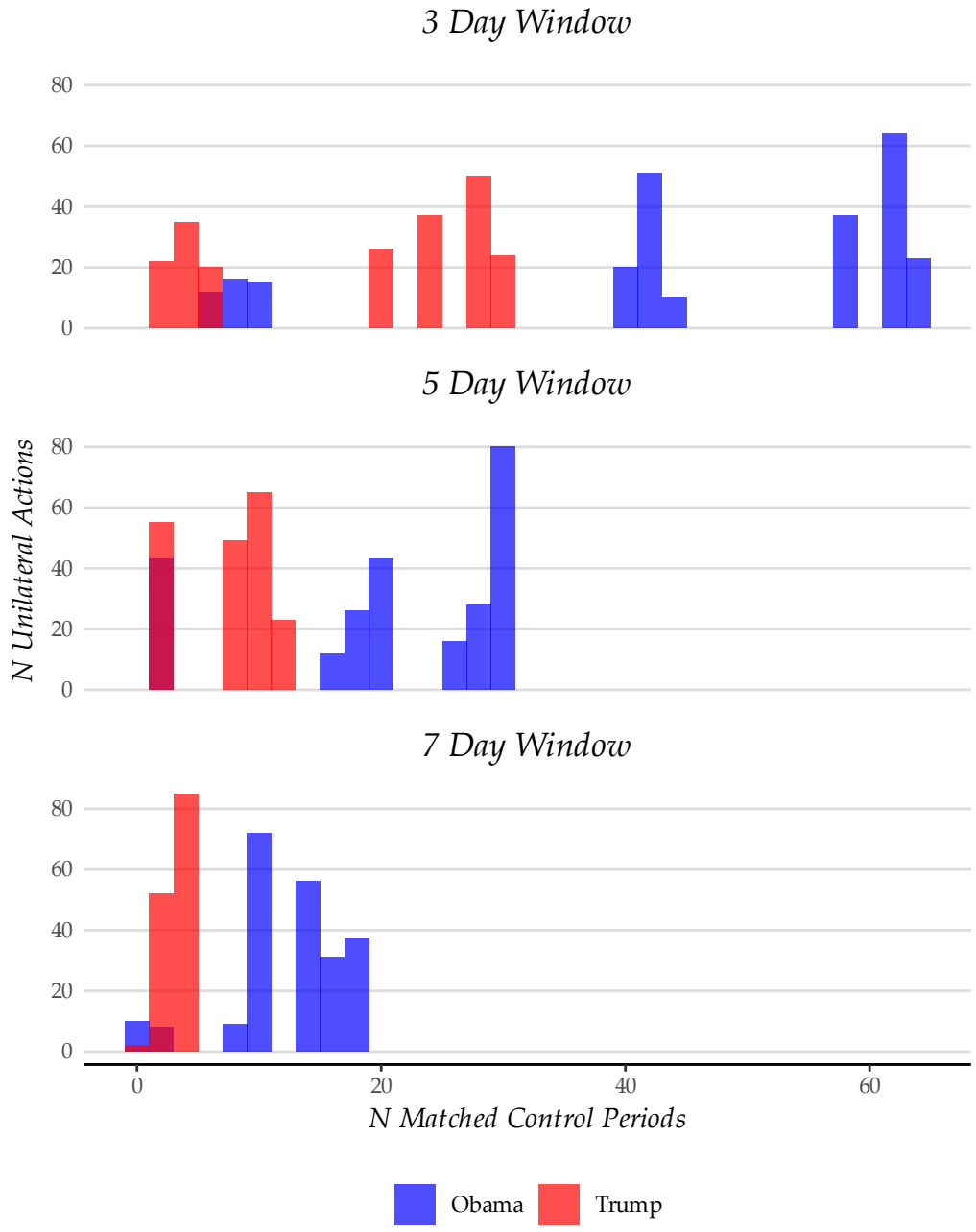


Table D.1 – Placebo Test Shows Mixed Evidence of Parallel Trends. Replicates the results in Table 1 but sets the date of treatment to 3 days before the issuance of unilateral actions. As such, the results test whether searches for the president increase before the eventual issuance of a unilateral action.

	Pooled	Trump	Obama
Post	-0.01^{***}	-0.01^{***}	-0.00
	(0.00)	(0.00)	(0.00)
Treatment	0.13	0.16	0.10
	(0.07)	(0.12)	(0.08)
Post*Treatment	0.11	0.17	0.06
	(0.06)	(0.11)	(0.06)
Action FE	✓	✓	✓
Division FE	✓	✓	✓

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Figure D.2 – Number of Matched Control Units Over Time. Plots the total number of control units by month. Control units computed using region-level Google Trends search results.

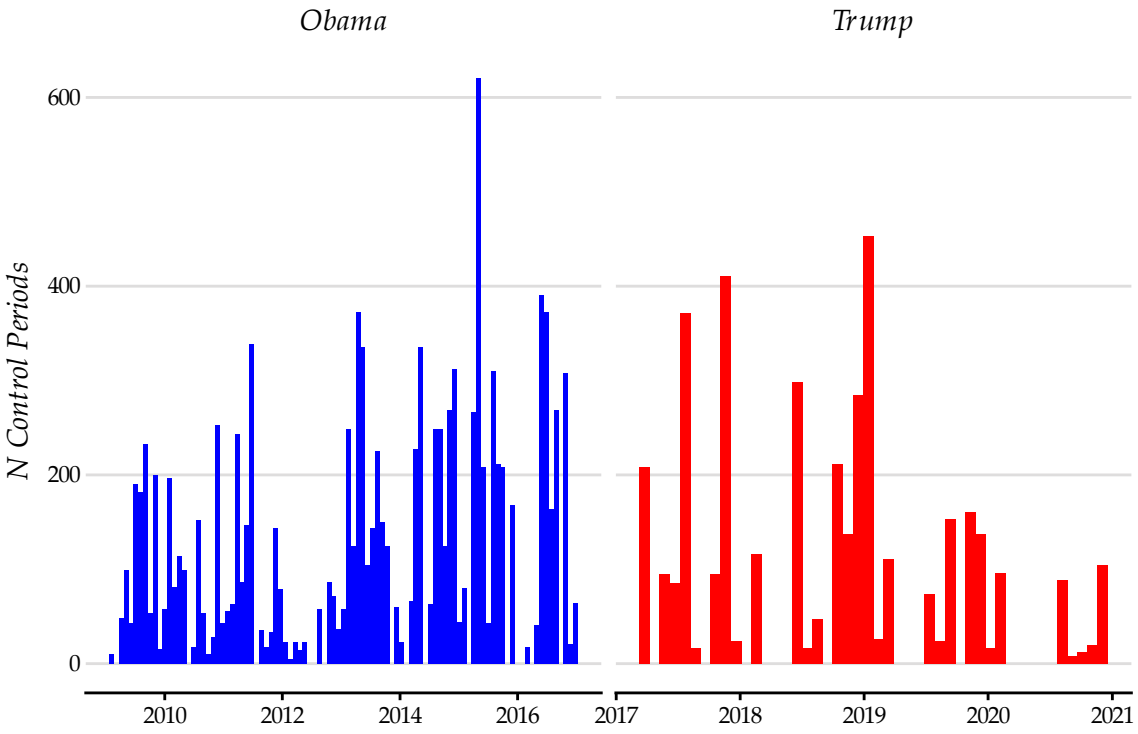


Figure D.3 – Share of Actions Matched by Year. Shows the share of unilateral actions in our dataset successfully matched to at least one control period per year. Actions unable to be matched to a control period or unable to be matched to a control period with any national search results comprise the remainder. Blue shading indicates Obama presidency and the red shading denotes Trump's first term.

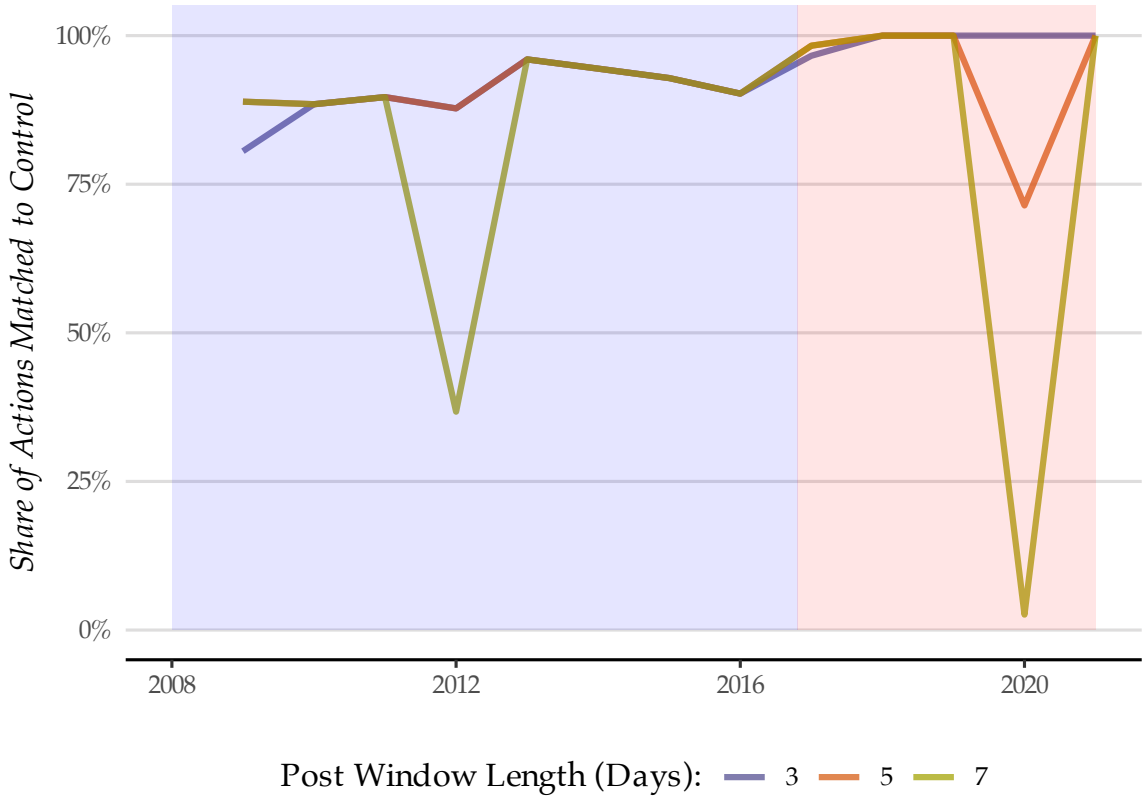
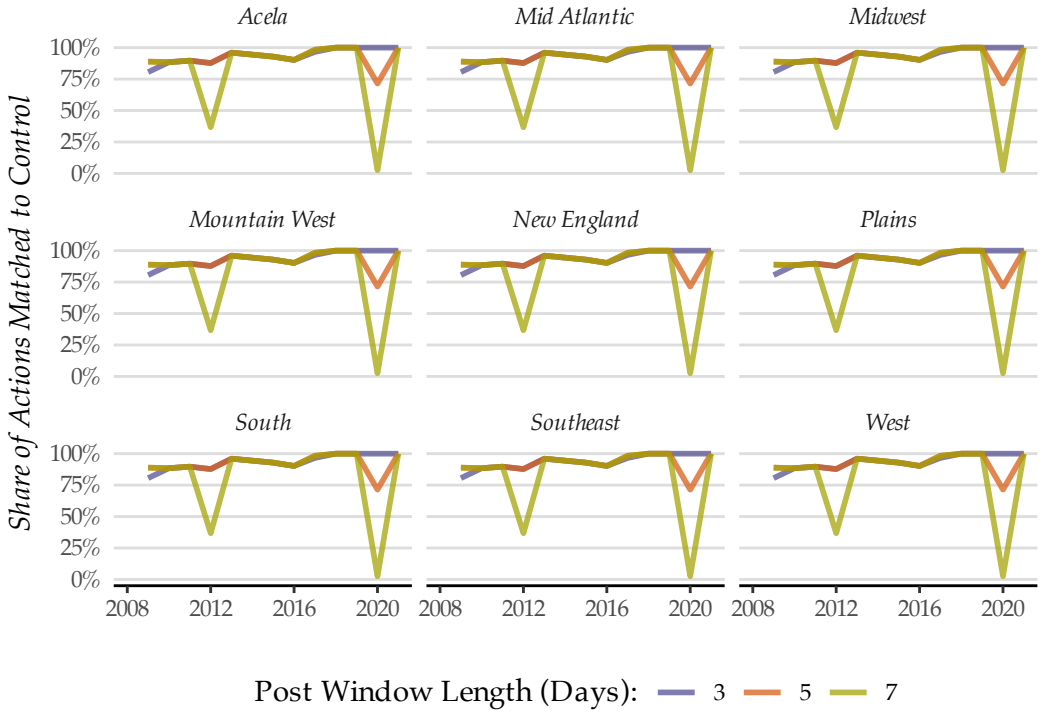


Figure D.4 – Share of Actions Matched, by Region and Year. Shows the share of unilateral actions in our dataset successfully matched to at least one control period per year and region. Actions unable to be matched to a control period or unable to be matched to a control period with any national search results comprise the remainder.



E Additional Results

We can also show our results with national-level action significance scores and nationally-aggregated search results, which removes the issue with low search results for some regions in Obama's first term. Figure E.1 shows these findings. In this setup, actions lead to increased searches up to 7 days following issuance when pooled across presidents. Although not statistically distinguishable, searches for Obama appear more robust than those for Trump, perhaps reflecting increasingly short news cycles. Overall, however, the results mirror those in Figure 6. Executive action that results in subsequent news coverage leads to a measurable shift in public attention toward the incumbent.

Figure E.1 – Main Results Largely Hold Using National Search Results. Replicates Figure 6 except the search results are measured at the national rather than regional level. Figure D.1 shows the distributions of control units by interval length and president. Full tabular results in Tables E.1, E.2, and E.3.

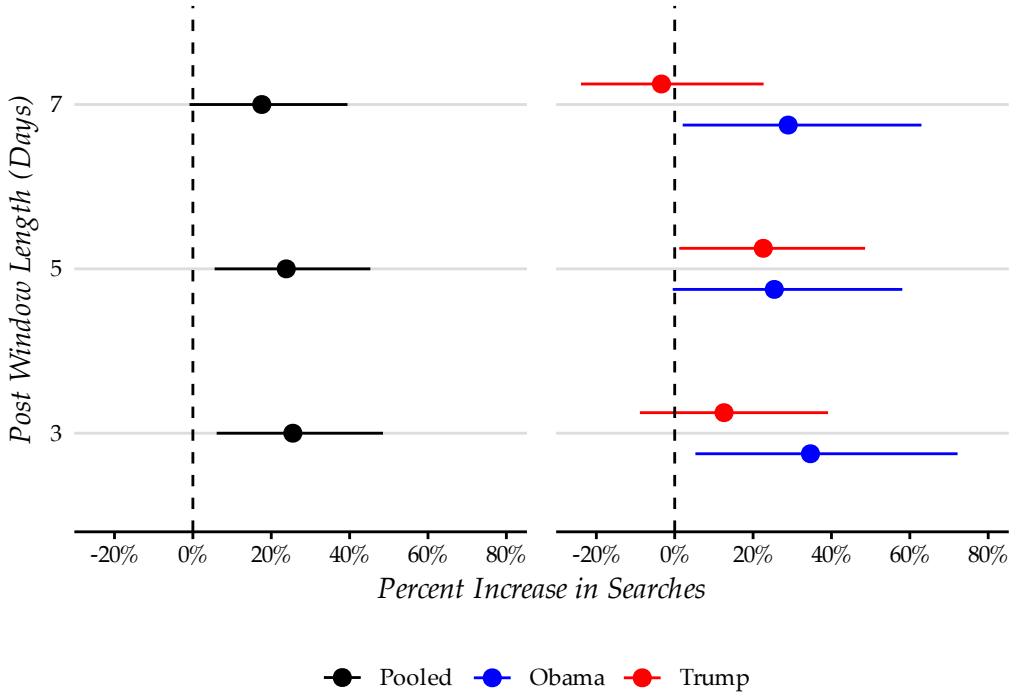


Table E.1 – Effect of Unilateral Action on National Searches for President (3 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of national Google searches for the given president in a 3 day period.

	Pooled	Trump	Obama
Post	-0.40^{***}	-0.37^{***}	-0.41^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.29^{***}	0.40^{**}	0.21^{*}
	(0.09)	(0.15)	(0.10)
Post*Treatment	0.23^{***}	0.12	0.30^{***}
	(0.05)	(0.07)	(0.08)
Action FE	✓	✓	✓
N Actions	462	214	248

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Figure E.2 – Effects of Unilateral Actions Isolated to Actions Covered by Newspapers. Shows the results of an alternative set of tests where the treatment variable is whether the unilateral action received any coverage by a newspaper. Full results in Tables E.10, E.11, and E.12.

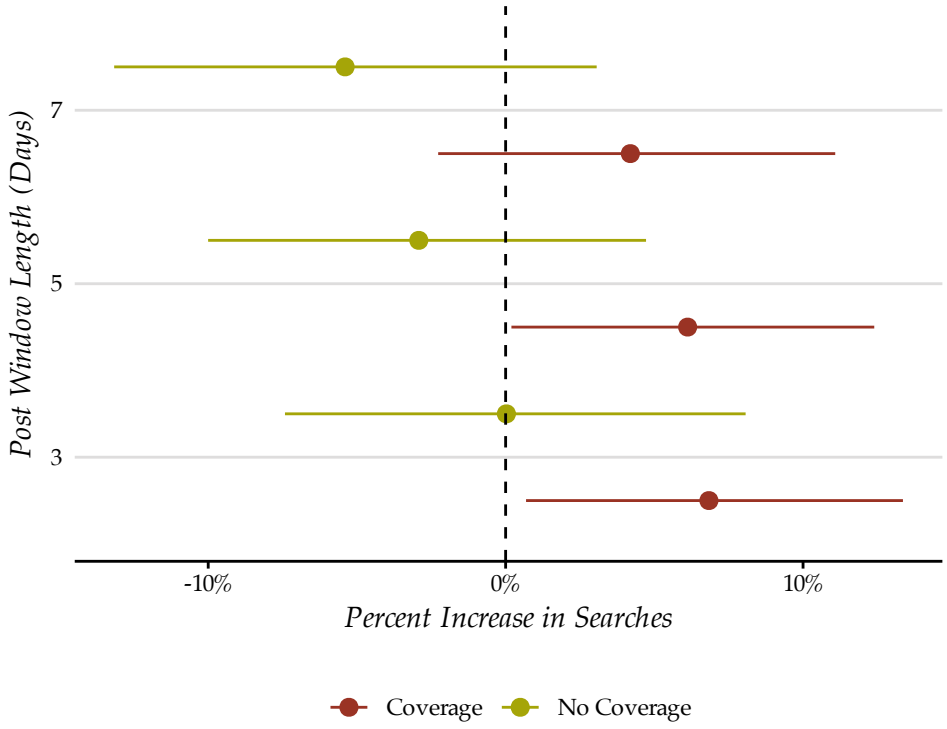


Table E.2 – Effect of Unilateral Action on National Searches for President (5 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of national Google searches for the given president in a 5 day period.

	Pooled	Trump	Obama
Post	0.29^{***}	0.29^{***}	0.29^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.34^{***}	0.41^{**}	0.28^{**}
	(0.09)	(0.15)	(0.11)
Post*Treatment	0.21^{***}	0.20^{*}	0.23^{**}
	(0.05)	(0.08)	(0.07)
Action FE	✓	✓	✓
N Actions	440	192	248

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.3 – Effect of Unilateral Action on National Searches for President (7 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of national Google searches for the given president in a 7 day period.

	Pooled	Trump	Obama
Post	0.70^{***}	0.75^{***}	0.69^{***}
	(0.00)	(0.01)	(0.00)
Treatment	0.46^{***}	0.52^{**}	0.45^{***}
	(0.09)	(0.16)	(0.11)
Post*Treatment	0.16^{**}	-0.03	0.25^{**}
	(0.06)	(0.07)	(0.08)
Action FE	✓	✓	✓
N Actions	362	139	223

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.4 – Effect of Unilateral Actions on Searches for the President (3 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 3 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are dropped from the analysis. Unilateral action and regional fixed effects are included and standard errors are clustered by month.

	Pooled	Trump	Obama
Post	-0.40^{***}	-0.37^{***}	-0.40^{***}
	(0.01)	(0.03)	(0.01)
Treatment	0.26	0.37	0.17
	(0.20)	(0.34)	(0.14)
Post*Treatment	0.22^{***}	0.11	0.28^{**}
	(0.06)	(0.11)	(0.09)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	4122	1917	2205

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.5 – Effect of Unilateral Actions on Searches for the President (3 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 3 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are set to the minimum number of searches for the given president. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

	Pooled	Trump	Obama
Post	-0.40^{***}	-0.37^{***}	-0.40^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.13	0.32^{*}	-0.03
	(0.10)	(0.14)	(0.14)
Post*Treatment	0.23^{***}	0.12	0.30^{***}
	(0.05)	(0.07)	(0.07)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	4158	1926	2232

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.6 – Effect of Unilateral Actions on Searches for the President (5 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 5 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are set to the minimum number of searches for the given president. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

	Pooled	Trump	Obama
Post	0.29^{***}	0.29^{***}	0.29^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.22^{*}	0.34^{*}	0.12
	(0.09)	(0.14)	(0.11)
Post*Treatment	0.28^{***}	0.23^{*}	0.32^{***}
	(0.06)	(0.09)	(0.09)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	3960	1728	2232

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.7 – Effect of Unilateral Actions on Searches for the President (5 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 5 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are dropped from the analysis. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

	Pooled	Trump	Obama
Post	0.29^{***}	0.29^{***}	0.29^{***}
	(0.00)	(0.00)	(0.00)
Treatment	0.30^{***}	0.38^{**}	0.23^{*}
	(0.08)	(0.14)	(0.09)
Post*Treatment	0.20^{***}	0.19^{*}	0.21^{**}
	(0.05)	(0.08)	(0.07)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	3960	1728	2232

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.8 – Effect of Unilateral Actions on Searches for the President (7 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 7 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are set to the minimum number of searches for the given president. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

	Pooled	Trump	Obama
Post	0.70^{***}	0.76^{***}	0.69^{***}
	(0.00)	(0.01)	(0.00)
Treatment	0.32^{***}	0.47^{**}	0.25^{*}
	(0.09)	(0.15)	(0.11)
Post*Treatment	0.23^{**}	-0.03	0.35^{**}
	(0.08)	(0.09)	(0.12)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	3258	1251	2007

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.9 – Effect of Unilateral Actions on Searches for the President (7 day interval). Displays the results of three OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 7 day period. Regions are defined using custom divisions (N = 9). Control periods with an insufficient number of search results are dropped from the analysis. Unilateral action and regional fixed effects are included and standard errors are clustered by action.

	Pooled	Trump	Obama
Post	0.70^{***}	0.76^{***}	0.69^{***}
	(0.00)	(0.01)	(0.00)
Treatment	0.41^{***}	0.52^{***}	0.36^{***}
	(0.08)	(0.14)	(0.09)
Post*Treatment	0.14^{**}	-0.06	0.23^{**}
	(0.05)	(0.07)	(0.07)
Action FE	✓	✓	✓
Division FE	✓	✓	✓
N Region-Action Obs	3258	1251	2007

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.10 – Effect of Unilateral Actions on National Searches for the President, by Coverage Status (3 day window). Displays the results of six OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 3 day period. Unilateral action fixed effects are included.

	Pooled		Trump		Obama	
	Coverage	No coverage	Coverage	No coverage	Coverage	No coverage
Post	-0.39^{***}	-0.40^{***}	-0.37^{***}	-0.38^{***}	-0.40^{***}	-0.41^{***}
	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)
Treatment	0.08^{**}	0.03	0.10^{*}	0.02	0.07	0.05
	(0.03)	(0.04)	(0.04)	(0.05)	(0.04)	(0.05)
Post*Treatment	0.07^{***}	0.00	0.02	-0.01	0.10^{***}	-0.01
	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Action FE	✓	✓	✓	✓	✓	✓
N Actions	287	175	139	75	148	100

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.11 – Effect of Unilateral Actions on National Searches for the President, by Coverage Status (5 day window). Displays the results of six OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 5 day period. Unilateral action fixed effects are included.

	Pooled		Trump		Obama	
	Coverage	No coverage	Coverage	No coverage	Coverage	No coverage
Post	0.29^{***}	0.29^{***}	0.30^{***}	0.28^{***}	0.29^{***}	0.29^{***}
	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)
Treatment	0.10^{**}	0.05	0.11^{*}	0.03	0.10^{*}	0.05
	(0.03)	(0.04)	(0.05)	(0.06)	(0.04)	(0.05)
Post*Treatment	0.06^{***}	-0.03	0.05^{*}	-0.01	0.06^{*}	-0.04
	(0.02)	(0.02)	(0.03)	(0.04)	(0.03)	(0.03)
Action FE	✓	✓	✓	✓	✓	✓
N Actions	276	164	128	64	148	100

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table E.12 – Effect of Unilateral Actions on National Searches for the President, by Coverage Status (7 day window). Displays the results of six OLS models implementing our matched difference-in-differences design. The outcome variable is the logged number of Google searches for the given president in a 7 day period. Unilateral action fixed effects are included.

	Pooled		Trump		Obama	
	Coverage	No coverage	Coverage	No coverage	Coverage	No coverage
Post	0.71***	0.70***	0.76***	0.76***	0.69***	0.70***
	(0.00)	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)
Treatment	0.15***	0.09*	0.15**	0.08	0.16***	0.11
	(0.03)	(0.05)	(0.06)	(0.07)	(0.04)	(0.06)
Post*Treatment	0.04*	−0.06	−0.04	−0.13*	0.07*	−0.04
	(0.02)	(0.03)	(0.03)	(0.06)	(0.03)	(0.03)
Action FE	✓	✓	✓	✓	✓	✓
N Actions	235	127	98	41	137	86

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Figure E.3 – National Results (Forward Imputation of Search Results). Replicates the results of our main analysis in Figure E.1 using forward-imputed Google Trends search results.

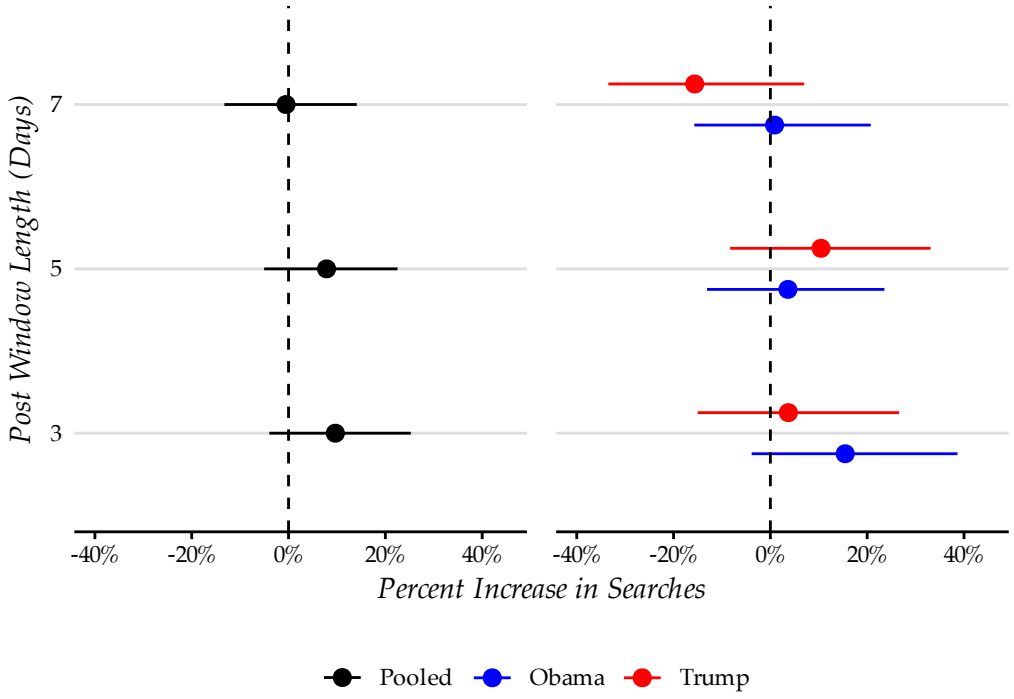
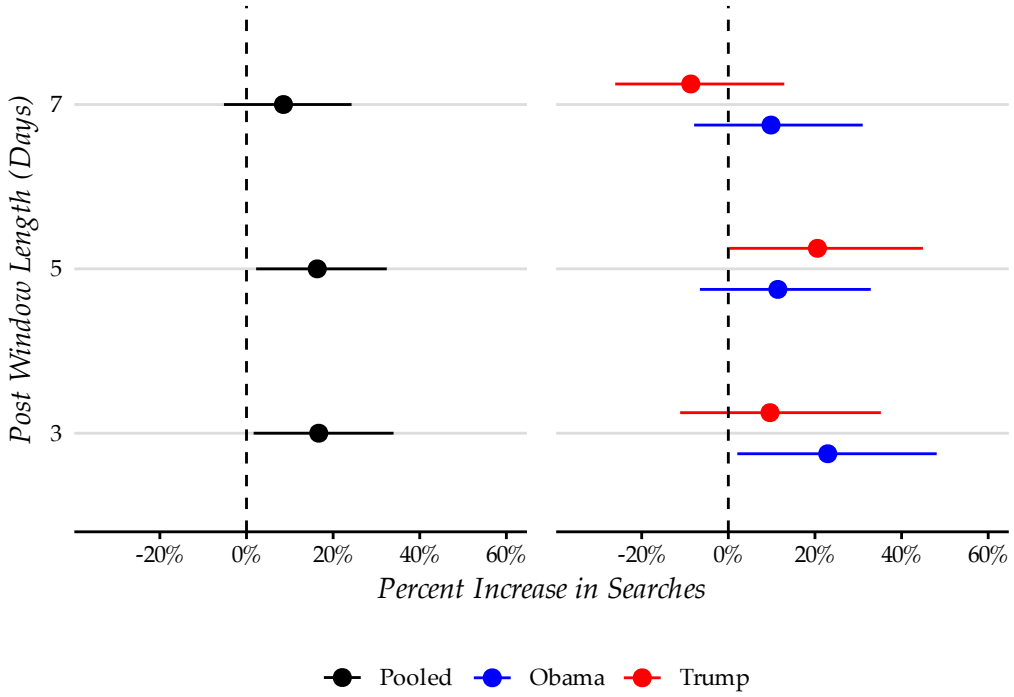


Figure E.4 – National Results, Excluding Actions Issued in 1st Month (Forward Imputation of Search Results). Replicates the results of our main analysis in Figure E.1 using forward-imputed Google Trends search results. Unlike Figure E.3, actions issued in the first month of the Obama and Trump administrations are dropped.



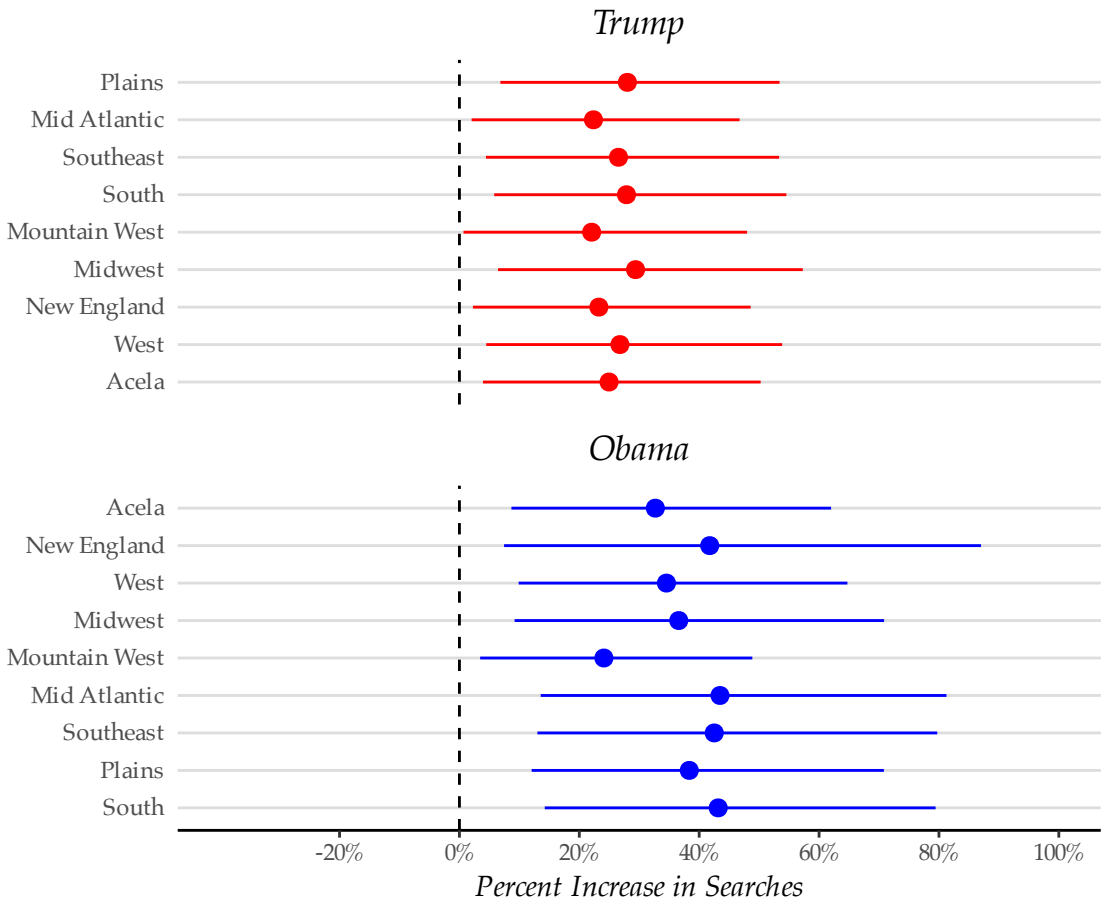
F Regional Partisanship Results

The descriptive evidence in Figure 7 is consistent with broader arguments about the nationalization of U.S. politics and media. Past scholarship, of course, has given us reason to believe this might be the case. Hopkins (2018) shows that the appetite for state and local news has declined. In addition, Peterson (2021) finds that over this time period, staff cuts have led to a decline in region-specific political stories.

The fact that there is little regional variation in news salience suggests that there may be little variation in attention attributable to (regional) politics. Newspapers should be anticipating what their readers will be interested in, and cover actions relevant to those readers more, which should increase public attention. Nonetheless, none of the preceding actually assesses whether there is variation in attention spikes associated with regions' political leanings. To do that, we repeat our analysis by region and president.

We plot the results in Figure F.1, which sorts each coefficient in order of regional support, so that the top region is the set of states with the high proportion supporting the incumbent in the last election. In general, Figure F.1 shows little evidence of regional variation. Although the estimates are noisy, there is no clear trend of more or less supportive regions seeking out more information about the president after a unilateral action.

Figure F.1 – Regional Results, Sorted by Presidential Support. Shows our main results estimated separately by region. The estimates are sorted based on the share of voters in the region who voted for Trump in 2016 and Obama in 2012, respectively. The interval length is set to 5 days.



This evidence, however, is merely speculative. Many things differ across regions, and the effects are mostly similar. This, of course, does not mean that asymmetric attention effects are not present. It might simply be that our aggregate search traffic data are ill suited to capturing them. Regional partisan lean may simply be too coarse a measure. What is likely needed is individual-level opinion and web-tracking data.